

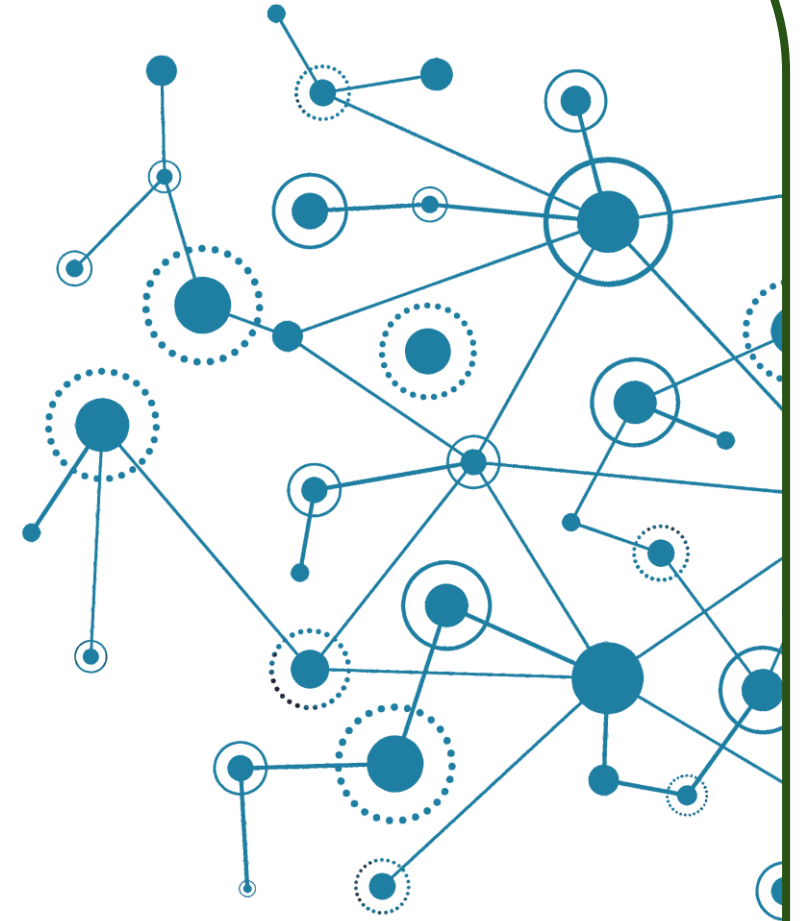
Go Hack Yourself. Evolving Tech Skills for the Age of AI & Automation

Ron Woerner, CISSP, CISM

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Cyber AAA, LLC

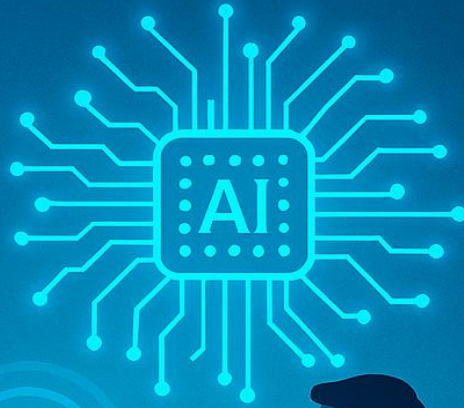
<https://www.linkedin.com/in/ronwoerner/>

Slides are available at: <https://github.com/hackerron/AI-Cybersecurity/blob/main/RWoerner%20-%20Evolving%20Cyber%20Skills%20-%20Sept2025.pdf>



Hack Yourself!

Evolving Tech Skills for the Age of AI & Automation



Learning objectives:

1. Recognize and Prioritize Emerging Tech Skills
2. Bridge the Gap Between Strategic and Technical Thinking
3. Develop a Personal and Organizational Upskilling Strategy
4. Integrate Cross-Disciplinary Perspectives into Practice

Exploring the strategic and practical dimensions of AI in technology.

>whoami

Ron Woerner

- Hacker
- Cybersecurity Consultant / Trusted Advisor
- Adjunct Professor, Bellevue University
- 25+ years experience in IT / Security
- Blogger, writer, and podcaster

LinkedIn:

<https://www.linkedin.com/in/ronwoerner/>



TEDx Omaha, 2019,
[Hackers Wanted](#)

Websites & Social Media:

<https://linktr.ee/cyberron>

A person wearing a dark hoodie is shown in profile, looking upwards. The background is solid black, and the lighting highlights the texture of the hoodie and the person's face.

WHY
are we here?

Warning!

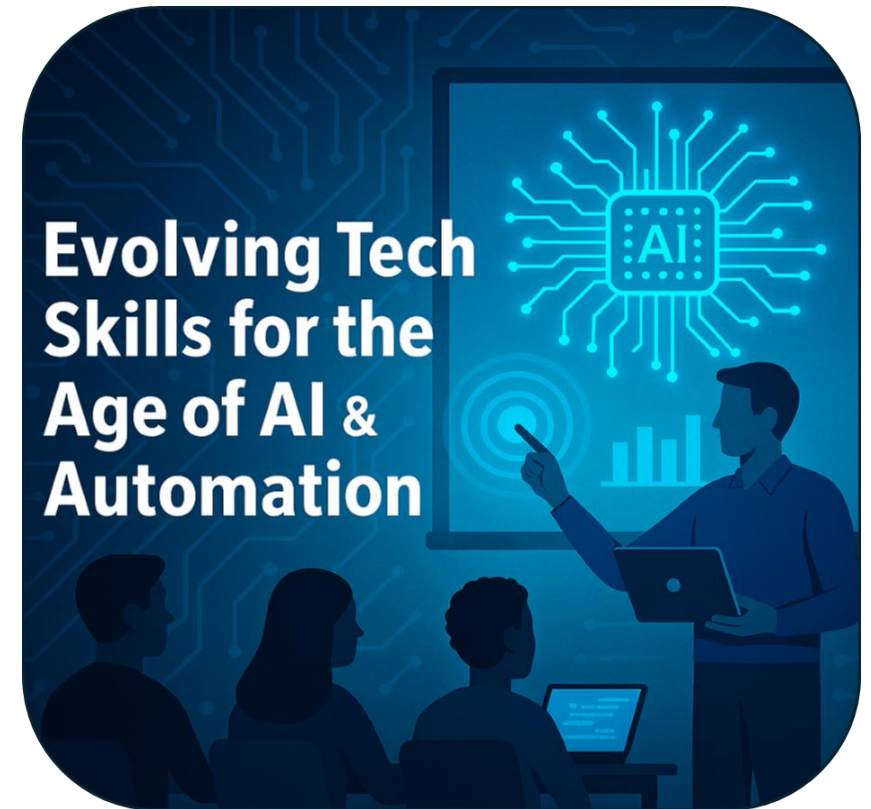
- These are my thoughts based on my studies and experiences and NOT necessarily those of my employers or anyone else
- Ethical uses only
- Use at your own risk / Normal caveats apply
- This talk goes FAST!
- There is homework!



What are you doing to enhance your AI & tech knowledge & skills?

Group Discussion

- Training resources
- Tools & apps
- Other?



Agenda

Setting the Stage

Home Labs: Your Personal Launchpad

AI Skill Building

AI & Security

Turning Skills into Career Momentum



MYTH: *“I need expensive gear to get started”*

Reality: *“Curiosity + structure beats hardware every time”*



AI generated image

Build a Resilient Mindset

Build a Resilient Mindset

BUILD A RESILIENT MINDSET

Build a Resilient Mindset

Build a Resilient Mindset

A-B-C = Always Be Curious



Building Skills: Experiment with Hacking Techniques*

- Use *FREE* frameworks & resources
- *Ethically & safely.*
Set up a home lab.
- Learn through doing
- Ask questions
- Sun Tzu's The Art of War:
"Know the enemy and know yourself; in a hundred battles you will never be in peril."

** More coming up later*



Frameworks, Standards, & Protocols

Name some common frameworks,
standards, and protocols
[NOT tools]



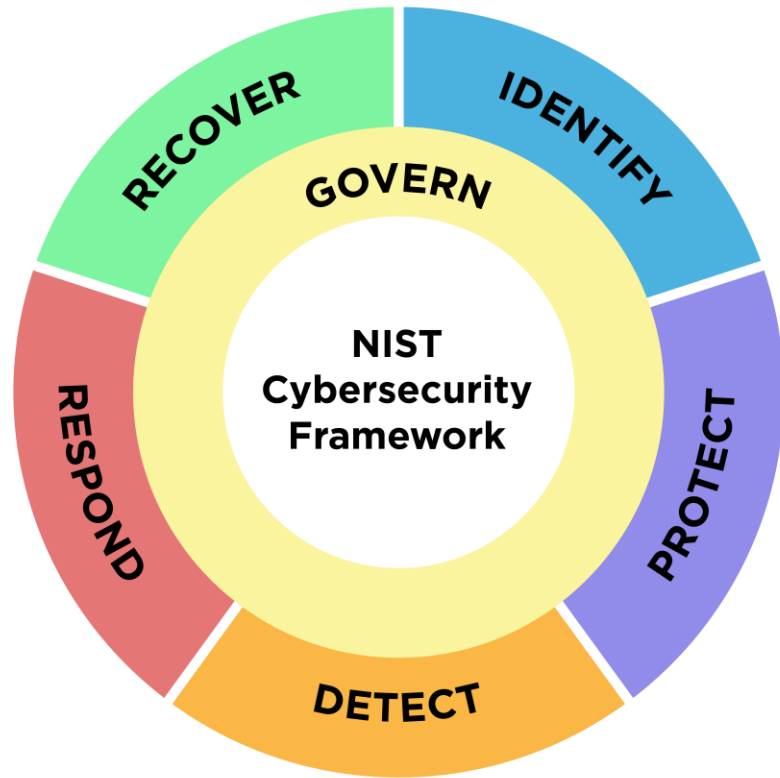
NIST – Computer Security Resource Center

<https://csrc.nist.gov/>

The **Computer Security Resource Center (CSRC)** has information on many of NIST's cybersecurity- and information security-related projects, publications, news and events. CSRC supports people and organizations in government, industry, and academia—both in the U.S. and internationally.

- Learn more about [current projects](#) and [upcoming events](#)
- Search and browse our [publications library](#) of current and historical standards, guidelines, and other reports
- Explore content by [topic](#)
- Search our [glossary](#) of terms defined in our publications

What does NIST Provide?



<https://www.nist.gov/cyberframework>

<https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>

The screenshot shows the NIST CSRC website. The header includes the NIST logo, "Information Technology Laboratory", "COMPUTER SECURITY RESOURCE CENTER", and the CSRC logo. A search bar and "CSRC MENU" are also visible. The main content area features a sidebar with links to "Projects", "Publications", "Topics", "News & Updates", and "Events". The main section is titled "Glossary" and includes a list of letters (A-Z) for navigation. Below the letters, it says "Sorted By: X Term (A-Z)" and "Showing 1 through 100 of 7970 matching records." At the bottom, there is a pagination link: <https://csrc.nist.gov/glossary>.

NIST CSF, PF, & RMF Goals

FREE!

- **Consistent** and **cost-effective** application of security controls
- **Repeatable** processes and assessment results
- Based on **existing standards**, guidelines, and practices
- **Voluntary**
- **Technology neutral and flexible**
- **Used across sectors**
- **Understanding** of enterprise-wide mission **risks**
- **Implement** an **efficient risk-based** security and privacy program

NIST CSF v2.0

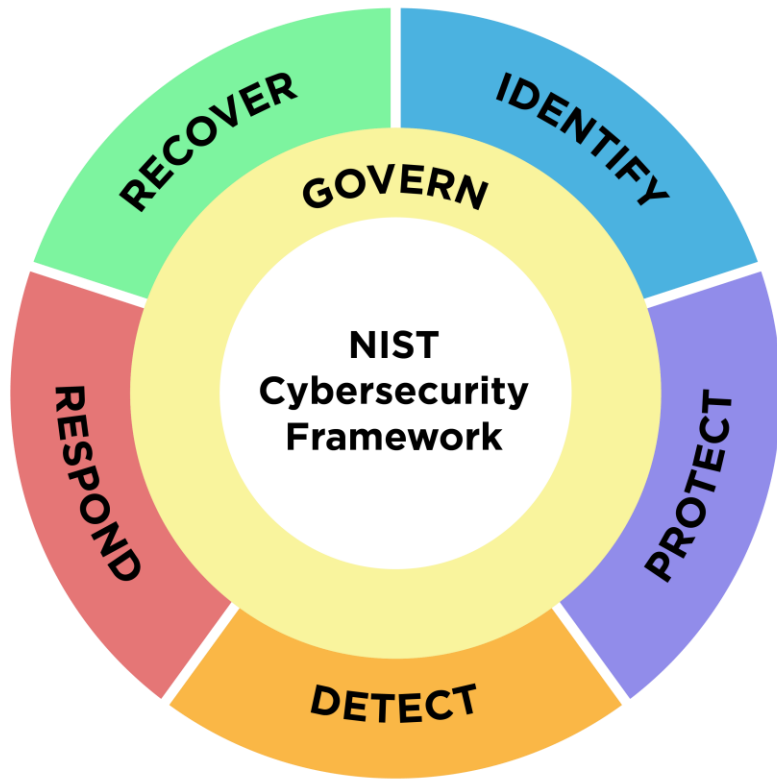


Table 1. CSF 2.0 Core Function and Category names and identifiers

Function	Category	Category Identifier
<u>Govern (GV)</u>	Organizational Context	GV.OC
	Risk Management Strategy	GV.RM
	Roles, Responsibilities, and Authorities	GV.RR
	Policy	GV.PO
	Oversight	GV.OV
	Cybersecurity Supply Chain Risk Management	GV.SC
<u>Identify (ID)</u>	Asset Management	ID.AM
	Risk Assessment	ID.RA
	Improvement	ID.IM
<u>Protect (PR)</u>	Identity Management, Authentication, and Access Control	PR.AA
	Awareness and Training	PR.AT
	Data Security	PR.DS
	Platform Security	PR.PS
	Technology Infrastructure Resilience	PR.IR
<u>Detect (DE)</u>	Continuous Monitoring	DE.CM
	Adverse Event Analysis	DE.AE
<u>Respond (RS)</u>	Incident Management	RS.MA
	Incident Analysis	RS.AN
	Incident Response Reporting and Communication	RS.CO
	Incident Mitigation	RS.MI
<u>Recover (RC)</u>	Incident Recovery Plan Execution	RC.RP
	Incident Recovery Communication	RC.CO

<https://www.nist.gov/cyberframework>

<https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>

Center for Internet Security

<https://www.cisecurity.org/controls>



CONTROL 01 Inventory and Control of Enterprise Assets 5 Safeguards IG1 2/5 IG2 4/5 IG3 5/5	CONTROL 02 Inventory and Control of Software Assets 7 Safeguards IG1 3/7 IG2 6/7 IG3 7/7	CONTROL 03 Data Protection 14 Safeguards IG1 6/14 IG2 12/14 IG3 14/14
CONTROL 04 Secure Configuration of Enterprise Assets and Software 12 Safeguards IG1 7/12 IG2 11/12 IG3 12/12	CONTROL 05 Account Management 6 Safeguards IG1 4/6 IG2 6/6 IG3 6/6	CONTROL 06 Access Control Management 8 Safeguards IG1 5/8 IG2 7/8 IG3 8/8
CONTROL 07 Continuous Vulnerability Management 7 Safeguards IG1 4/7 IG2 7/7 IG3 7/7	CONTROL 08 Audit Log Management 12 Safeguards IG1 3/12 IG2 11/12 IG3 12/12	CONTROL 09 Email and Web Browser Protections 7 Safeguards IG1 2/7 IG2 6/7 IG3 7/7
CONTROL 10 Malware Defenses 7 Safeguards IG1 3/7 IG2 7/7 IG3 7/7	CONTROL 11 Data Recovery 5 Safeguards IG1 4/5 IG2 5/5 IG3 5/5	CONTROL 12 Network Infrastructure Management 8 Safeguards IG1 1/8 IG2 7/8 IG3 8/8
CONTROL 13 Network Monitoring and Defense 11 Safeguards IG1 0/11 IG2 6/11 IG3 11/11	CONTROL 14 Security Awareness and Skills Training 9 Safeguards IG1 8/9 IG2 9/9 IG3 9/9	CONTROL 15 Service Provider Management 7 Safeguards IG1 1/7 IG2 4/7 IG3 7/7
CONTROL 16 Applications Software Security 14 Safeguards IG1 0/14 IG2 11/14 IG3 14/14	CONTROL 17 Incident Response Management 9 Safeguards IG1 3/9 IG2 8/9 IG3 9/9	CONTROL 18 Penetration Testing 5 Safeguards IG1 0/5 IG2 3/5 IG3 5/5

Home Labs: Your Personal Launchpad



Before you do anything else...

Use what you have in front of you.

Perform an audit of your home computer set up

What's on your network? / Asset Inventory

- a. Hardware – IOT, Laptops, Tablets, Mobile, Network Routers
- b. OS versions
- c. Applications
- d. Data

Required for an AI architecture!

General Network Audit Steps

1. Review network configuration controls
2. Establish an application inventory
3. Vulnerability management – Update, Config, Compensating Controls
4. Disable unnecessary services
5. SNMP Management
6. Identity & Access Management
7. Authentication (passwords & MFA)
8. Verify secure management protocols
9. Backup configuration files

General Network Audit Steps

10. Network logging & monitoring – local and centralized (SIEM)
11. Use NTP
12. Manage administrator & privileged access (console admin)
13. Physical security of network equipment
14. Standard naming conventions
15. Verify standard, documented processes
16. Evaluate use of Network Access Controls (NAC) & Zero Trust
17. Remote access management (general & admin)
18. Review Disaster Recovery & Business Continuity Plans (DR/BCP)

You Try It – Network Evaluation

What's your IP Address?

Computer Networking and System Administration

1. Open a command prompt (Windows) or Terminal (Linux, OSX).
2. Type *ipconfig*
3. *You can also do this on your pocket PC (aka smartphone),
Under Settings > General > About (iPhone)*
4. Compare against <https://whatismyipaddress.com/>

[Download](#)[Reference Guide](#)[Book](#)[Docs](#)[Zenmap GUI](#)[In the Movies](#)

Downloading Nmap

Get the latest Nmap for your system:

- [Windows](#)
- [macOS](#)
- [Linux \(RPM\)](#)
- [Any other OS \(source code\)](#)

Older versions (and sometimes newer test releases) are available from the [Nmap release archive](#) (and really old ones are in [dist-old](#)). For the more security-paranoid (smart) users, GPG detached signatures and SHA-1 hashes for each release are available in the [sigs directory](#) ([verification instructions](#)). Before downloading, be sure to read the relevant sections for your platform from the [Nmap Install Guide](#). The most important changes (features, bugfixes, etc) in each Nmap version are described in the [Changelog](#). Using Nmap is covered in the [Reference Guide](#), and don't forget to read the other [available documentation](#), particularly the official book [Nmap Network Scanning](#)!

Nmap users are encouraged to subscribe to the *Nmap-hackers* mailing list. It is a low volume (7 posts in 2015), moderated list for the most important announcements about Nmap, Insecure.org, and related projects. You can join the 128,953 current subscribers (as of September 2017) by submitting your email address here:

(or subscribe with custom options from the [Nmap-hackers list info page](#))

You can also get updates by liking [Nmap on Facebook](#) or following us [@nmap on Twitter](#).

Nmap is distributed with source code under [custom license terms](#) similar to (and derived from) the GNU General Public License, as noted in the [copyright page](#).

Wireshark

The image shows the Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, capture, analysis, and display. A display filter bar shows "Apply a display filter ... <Ctrl-/>".

The main packet list table displays the following data:

No.	Time	Source	Destination	Protocol	Length	Info
58	3.025899	2600:8804:1c00:6920...	2a03:2880:f36d:90:f...	TCP	404	49801 → 443 [PSH, ACK] Seq=2118 Ack=1 Win=255 Len=330 [TCP PDU reas...
59	3.025919	2600:8804:1c00:6920...	2a03:2880:f36d:90:f...	TLSv1.2	91	Application Data
60	3.085174	2a03:2880:f36d:90:f...	2600:8804:1c00:6920...	TCP	74	443 → 49801 [ACK] Seq=1 Ack=2118 Win=4420 Len=0
61	3.085222	2a03:2880:f36d:90:f...	2600:8804:1c00:6920...	TCP	74	443 → 49801 [ACK] Seq=1 Ack=2448 Win=4420 Len=0
62	3.085222	2a03:2880:f36d:90:f...	2600:8804:1c00:6920...	TCP	74	443 → 49801 [ACK] Seq=1 Ack=2465 Win=4420 Len=0
63	3.289156	192.168.0.103	20.50.80.212	TCP	55	58684 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
64	3.289156	192.168.0.103	20.50.80.212	TCP	55	51363 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
65	3.398980	2600:8804:1c00:6920...	2a03:2880:f36d:90:f...	TCP	79	49801 → 443 [PSH, ACK] Seq=2465 Ack=1 Win=255 Len=5
66	3.399042	2600:8804:1c00:6920...	2a03:2880:f36d:90:f...	TCP	404	49801 → 443 [PSH, ACK] Seq=2470 Ack=1 Win=255 Len=330 [TCP PDU reas...
67	3.399056	2600:8804:1c00:6920...	2a03:2880:f36d:90:f...	TLSv1.2	91	Application Data
68	3.408779	20.50.80.212	192.168.0.103	TCP	66	443 → 58684 [ACK] Seq=1 Ack=2 Win=16382 Len=0 SLE=1 SRE=2
69	3.408853	20.50.80.212	192.168.0.103	TCP	66	443 → 51363 [ACK] Seq=1 Ack=2 Win=16384 Len=0 SLE=1 SRE=2
70	3.432859	2a03:2880:f36d:90:f...	2600:8804:1c00:6920...	TCP	74	443 → 49801 [ACK] Seq=1 Ack=2470 Win=4420 Len=0
71	3.432908	2a03:2880:f36d:90:f...	2600:8804:1c00:6920...	TCP	74	443 → 49801 [ACK] Seq=1 Ack=2800 Win=4420 Len=0

A tooltip "Presentation last saved: Just now" is visible over packet 60.

The packet details pane for packet 63 shows:

- > Frame 63: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface 0
- > Ethernet II, Src: AzureWaveTec_c0:77:aa (f8:54:f6:c0:77:aa), Dst: Commsco...
- > Internet Protocol Version 4, Src: 192.168.0.103, Dst: 20.50.80.212
- > Transmission Control Protocol, Src Port: 58684, Dst Port: 443, Seq: 1, Ac...

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```
0000 10 e1 77 3b a1 33 f8 54 f6 c0 77 aa 08 00 45 00  ..w;3-T ..w...E..
0010 00 29 fb c7 40 00 80 06 d8 f1 c0 a8 00 67 14 32  ..)@... ..g.2
0020 50 d4 e5 3c 01 bb f3 b6 3a 70 9b b0 72 b2 50 10  P<<... :p.r.P.
0030 00 ff 65 3d 00 00 00  ..e=...
```

The status bar at the bottom indicates: wireshark Wi-FiTVLRK3.pcapng | Packets: 90 · Dropped: 0 (0.0%) | Profile: Default



Steps to Build a Budget-Friendly Home Tech Lab

Define Your Purpose

- **Decide what you want to learn:** Linux, networking, cybersecurity, AI, cloud computing, etc.
- This helps prioritize hardware, software, and setup choices

Repurpose Old Equipment

- **Use old laptops or desktops** to run Linux, host servers, or simulate networks
- Look for deals on routers, switches, and servers from trusted refurb vendors

Set Up a Basic Network

- Use a reliable router and Ethernet switch to connect devices
- **Segment your lab** equipment from the rest of your network
- Assign static IPs for easier management and testing



Steps to Build a Budget-Friendly Home Tech Lab

Free & Open-Source S/W

- Set up a **virtual lab** using VirtualBox, Proxmox, MS Hyper-V
- Install **Linux** distros (Ubuntu, Debian, Zorin OS)
- Use **open-source tools** like Wireshark, Metasploit, Docker, and FreeNAS for hands-on projects

Leverage Cloud Free* Tiers

- Platforms like AWS, Azure, and Google Cloud offer free tiers* for experimentation.
- Great for learning cloud architecture, identity management, and scalable services

Document and Share Your Projects

- **Keep notes**, screenshots, and diagrams of your setups.
- **Build a portfolio** or **GitHub repo** to showcase your skills to employers

VM Apps

Ratings:

<https://learn.g2.com/free-virtual-machine-software>

Instructions:

<https://serveracademy.com/blog/how-to-build-an-it-lab/>

Best free virtual machine software	G2 Rating	Free plan	Paid plan
Azure Virtual Machines	4.4/5	Free trial	Available on request
XenServer	4.1/5	Free + licensed versions	Available on request
Google Compute Engine	4.5/5	Free tier/free credits (varies)	Available on request
Oracle VM	4.3/5	Free download	No (legacy/limited updates)
Proxmox VE	4.6/5	Free download	Subscriptions starting at 1,320/yr
QEMU	4.3/5	Free download	No (legacy/limited updates)
Scale Computing Platform	4.7/5	Free trial (varies)	Starting at \$189/user/yr
VirtualBox	4.5/5	Free download	Available on request
VMware ESXi	4.7/5	Free download	Available on request
VMware Fusion	4.4/5	Free download	Available on request
VMware vSphere	4.5/5	Free download	Available on request



Source: <https://gbhackers.com/linux-distros/>

Critical Tools

1. Command line
2. [Notepad++](#) – Text editor (latest version)
3. Python (<https://www.python.org/>) – Programming Language
 - Install Python – <https://www.python.org/downloads/>
 - [Make sure you install “pip” and include both in your system PATH.] Note
 - [Python Tutorial](#) (for devs)
 - [Python Documentation](#)
 - DeepLearning.AI – [AI Python for Beginners](#)

4. APIs (Application Programming Interfaces)

- Explanation:
 - Sets of protocols and tools that allow developers to integrate artificial intelligence functionalities into their applications without having to build complex AI models from scratch.
 - They act as a bridge, enabling different software systems to communicate and exchange data.
 - Essentially, AI APIs provide access to pre-built and pre-trained AI services hosted in the cloud.
- Get Your Own OpenAI API Key:
 1. Sign up at platform.openai.com
 2. Go to [API Keys](#) and create a new key
 3. You'll use this with your AI Agents[OpenAI API Dev Quick Start Guide](#)



Questions?

AI Skillz

“AI is no longer optional... but reckless adoption is worse than slow adoption”

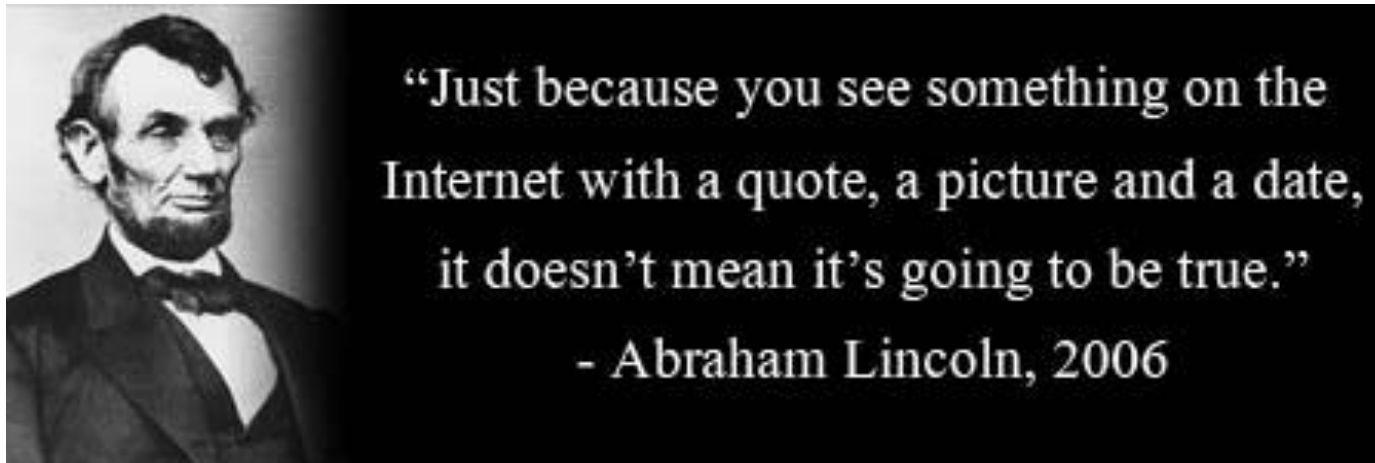
Jax Scott, https://www.linkedin.com/posts/iamjax_an-executives-beginner-guide-to-ai-without-activity-7422243300390215681-WUN_

AI, Tech, & Cyber Core Principle

Trust, AND Verify

“Never trust a single data point.”

from my life as a military intelligence officer



Quick animal fact #38



Less than 5% of ducks have attempted world domination

Mastering Fundamentals

AI:

- SDLC / Coding Best Practices
- Core Algorithms: Decision trees, neural networks, clustering
- Prompt Engineering

Cybersecurity:

- Computer Science, Math, Economics, Marketing, Psychology
- GRC
- Design Principles / Zero Trust
- Threat Modeling

Understanding the “why” behind the model helps you spot challenges and design more resilient systems.

Core AI Competencies for Cyber Professionals

Technical Skills to Cultivate:

AI Fundamentals:

Machine learning, deep learning, NLP, prompt engineering, and model evaluation

AI-Specific Threats:

Deepfakes, automated phishing, model inversion, data poisoning

Resilient Development & Deployment Practices:

Threat modeling, secure coding, and adversarial robustness

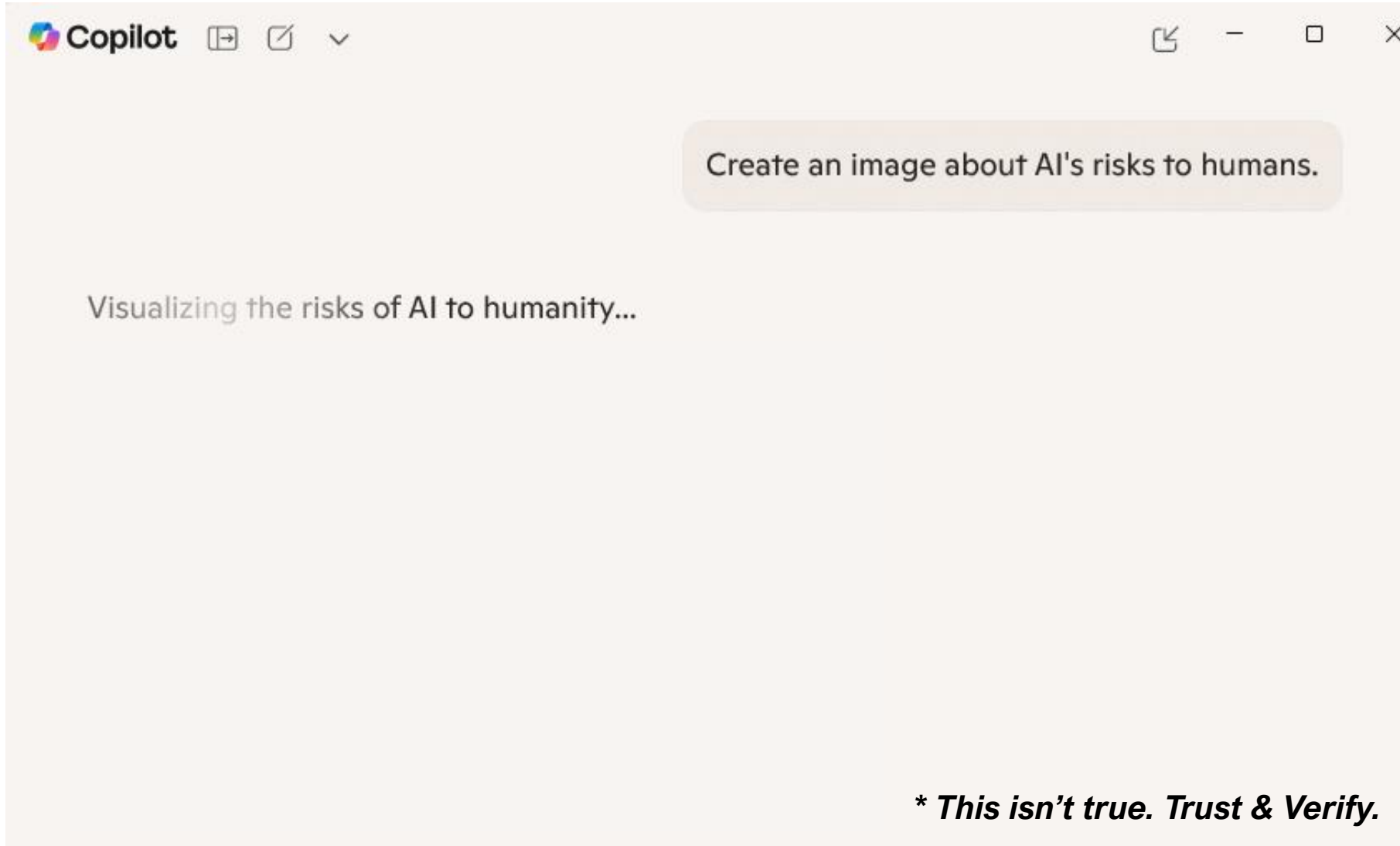
Data Handling & Privacy:

Governance, encryption, anonymization, and compliance (GDPR, CCPA)

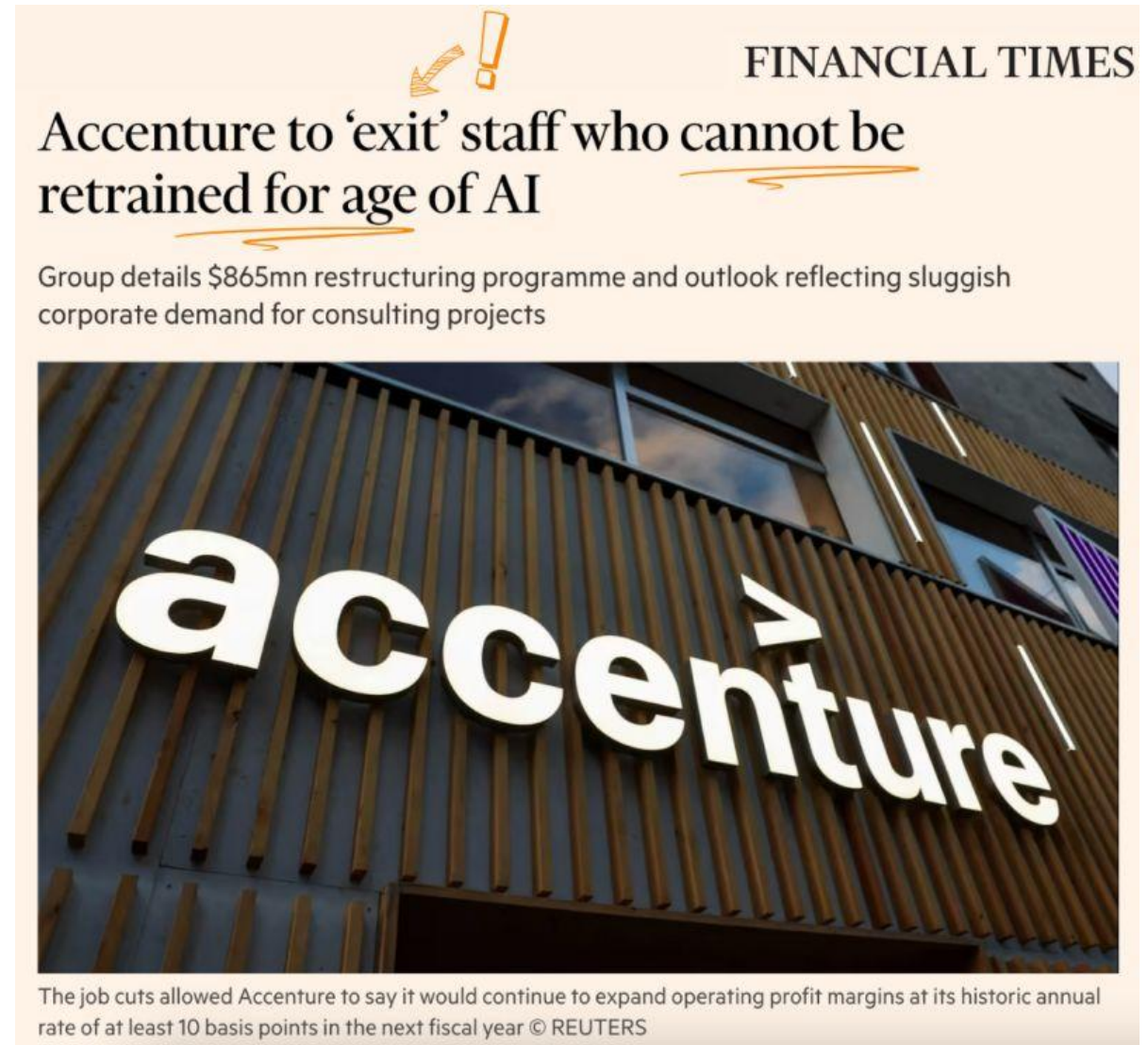
Security Architecture:

Identity management, container security, and Zero Trust architecture

AI Human Risks



AI Human Risks



<https://www.ft.com/content/a74f8564-ed5a-42e9-8fb3-d2bddb2b8675>

Philosophical Reflection

NIST AI Risk Management Framework

AI Risks & Trustworthiness

1. Valid and Reliable
2. Safe
3. Secure and Resilient
4. Accountable and Transparent
5. Explainable and Interpretable
6. Privacy-Enhanced
7. Fair – with Harmful Bias Managed

- How do you know what's “true” and verify results?
- Explore ethical dilemmas
 - What bias may be introduced?
 - What does it mean to “trust” a machine?
 - Can AI be “self-aware” of its misuse?
- Balancing automation with human involvement
- Embracing continuous learning and interdisciplinary fluency

NIST – AI Risks & Threats

NIST AI 100-1

AI RMF 1.0



Fig. 1. Examples of potential harms related to AI systems. Trustworthy AI systems and their responsible use can mitigate negative risks and contribute to benefits for people, organizations, and ecosystems.

NIST AI Risk Management Framework 1.0,
<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>, p.5

Ron Woerner – Evolving Cyber Skills

AI Terms & Definitions

Reference*:

<https://iapp.org/resources/article/key-terms-for-ai-governance/>

LLM (Large Language Model): A type of deep learning model trained on massive text datasets to understand and generate human-like language.

- **Core Use:** Text generation, summarization, translation, coding assistance
- **Examples:** GPT-4, Claude, Gemini, LLaMA

NLP (Natural Language Processing): An AI subfield that enables machines to understand, interpret, and generate human language.

- **Core Use:** Text classification, entity recognition, translation, summarization
- **Examples:** Siri, Alexa, ChatGPT, Google Translate

Multimodal Models: AI systems capable of processing and integrating information from multiple types of data — or “modalities” — such as text, images, audio, video, and sensor input.

GenAI (Generative AI): Models that create new content — text, images, audio, code — based on learned patterns from training data.

Agentic AI: Autonomous systems that can set goals, plan, and execute tasks with minimal human intervention. These systems use LLMs as a “brain” and orchestrate multiple agents to complete complex workflows.

Prompt: A question, instruction, or input given to an AI model to guide its response.

Vibe Coding: an emerging development style where programmers use natural language to guide AI tools in generating code.

- **Core Use:** Fast prototyping, solo app development, creative coding
- **Tools:** Replit, Cursor, GitHub Copilot

Agentic AI Explained

Agentic AI refers to artificial intelligence systems that can act independently to achieve specific goals. Unlike traditional AI that waits for instructions, agentic AI can:

- **Plan** what needs to be done
- **Act** without constant human input
- **Adapt** to changing situations
- **Collaborate** with other agents or humans

Think of it like giving an AI a mission, and it figures out how to complete it—step by step, using tools, making decisions, and even learning along the way.

Agentic AI vs AI Agents:

Agentic AI is the **overarching framework** that enables AI systems to **solve problems with minimal supervision**. Within this framework, *AI agents are the individual components* responsible for executing specific tasks autonomously.

While agentic AI understands user goals and orchestrates the problem-solving process, AI agents perform the tasks.

Quick Example:

Imagine a hospital using agentic AI to manage staff schedules:

- A nurse calls in sick.
- The agentic AI notices the gap, checks who's available, considers patient needs, and **automatically updates the schedule**.
- It might even message the replacement nurse and notify the team—**all without human intervention**.

This is more than automation—it's goal-driven decision-making.

AI Conversations: Prompting Fundamentals

Asking good questions



Clarity & Conciseness: Always key.



Context is Key: Emphasize why context matters.



Activity: Vague vs. specific prompts.



Specifying Format: How you want the output (list, email, table, etc.).



Persona & Tone: Telling Gemini who it is and how to respond.

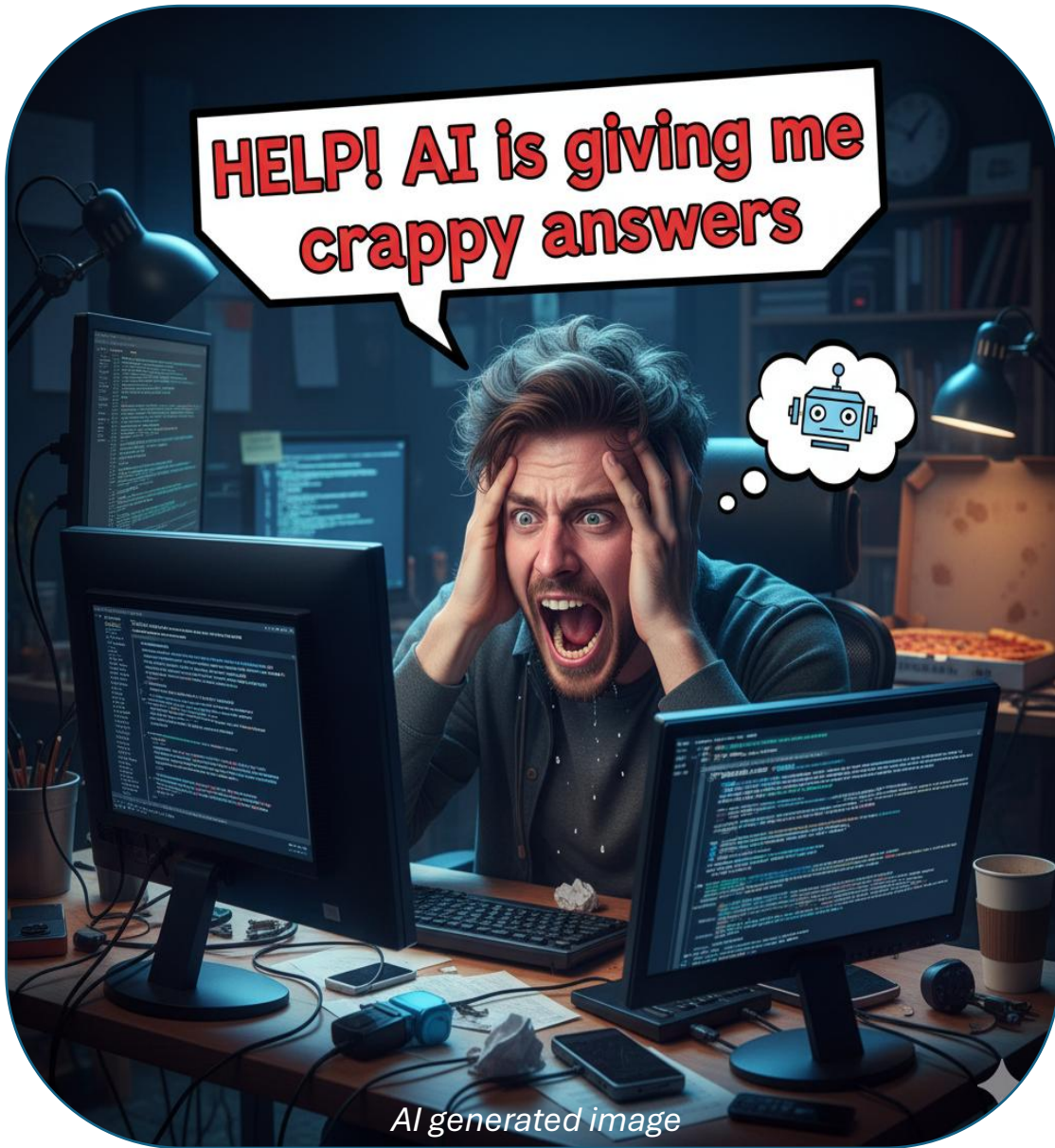


Constraints: What to avoid or include (e.g., "Keep it under 100 words," "Do not include personal names").



Iterate & Refine: AI is a dialogue; refine prompts for better results.

What are your prompting tips?



1. Prompt fundamentals
2. Ask it:
 - “Be direct, objective, expose blind spots, challenge assumptions”
 - Play “Devil’s Advocate.” Critique ideas from multiple personas & angles
 - For credible citations and flag weak evidence - Include source links
 - Show its work / “thought process”
 - For clarity and conciseness
3. Start over. Try a different tact

Experiment with AI Techniques

NEVER TEST IN PRODUCTION!!!

On the Open Internet

Experiment with AI Techniques

OSINT



ChatGPT



Gemini



perplexity



Hunter is your all-in-one email outreach platform. Find and connect with the people that matter to your business.



theHarvester

theHarvester is a simple to use, yet powerful tool designed to be used during the reconnaissance stage of a red team assessment or penetration test. It performs open source intelligence (OSINT) gathering to help determine a domain's external threat landscape. The tool gathers names, emails, IPs, subdomains, and URLs by using multiple public resources that include:

The first principle is that you must not fool yourself and you are the easiest person to fool.

Richard P. Feynman



Questions?

AI & Security

Security needs to be built in (OR ELSE!)

AI Cybersecurity Risks & Threats

Prompt: I'm building a presentation on cybersecurity and AI for a technical audience. Provide 5 ways ChatGPT and AI can be used maliciously.

Enhanced cyber reconnaissance

Automated social engineering

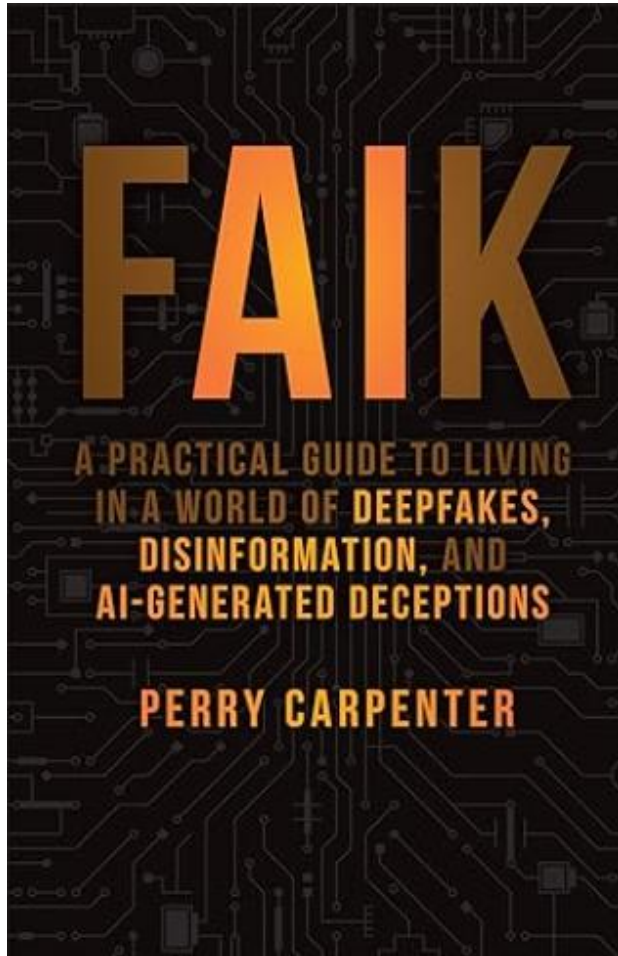
Supercharged phishing

Deepfakes and disinformation

Malware automation & mutation

Adversarial machine learning and data poisoning

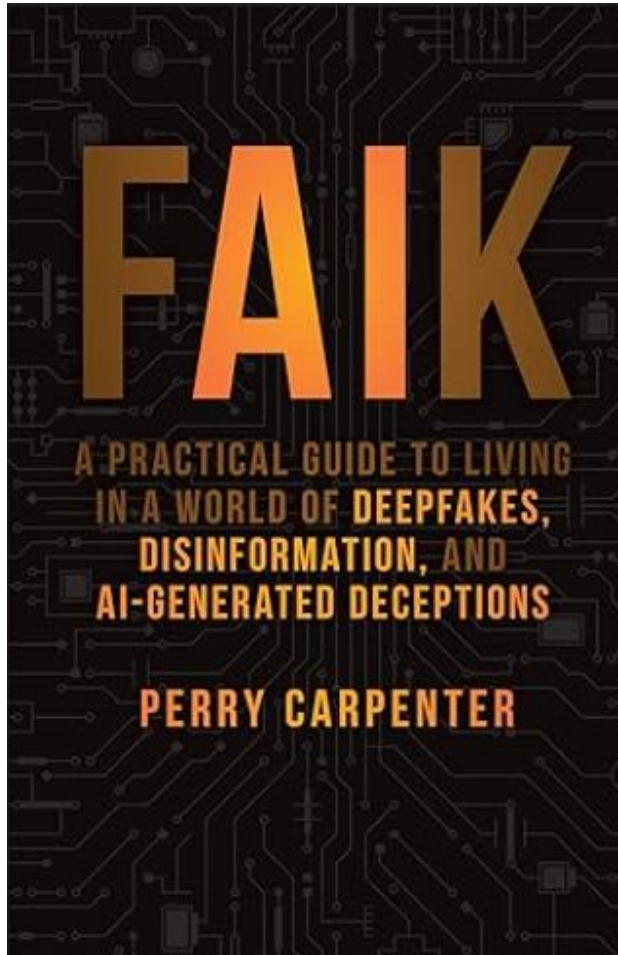
Deep Fakes



<https://www.amazon.com/FAIK-Practical-Disinformation-AI-Generated-Deceptions/dp/1394299885>



Deep Fakes



<https://www.amazon.com/FAIK-Practical-Disinformation-AI-Generated-Deceptions/dp/1394299885>



The FAIK Files

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Welcome to The FAIK Files! We explore the mysterious and often wacky intersection of AI, technology, and...more

thisbookisfaik.com and 4 more links

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For You



Deepfake SECRETS EXPOSED: Outsmart AI Deception with These Tricks!

6.1K views • 7 months ago



CREATE real-time DEEPFAKES (a.k.a. I became TAYLOR SWIFT...for Science!)

31K views • 8 months ago



Open Source, AI-Powered #FaikFiles Podcast 40.

814 views • 2 weeks ago

<https://www.youtube.com/@theFAIKfiles>

Top Questions about AI in Cybersecurity

- **GenAI in Security Tools:** Common use cases include incident summarization, threat research chatbots, and behavior modeling for triage and investigation.
- **AI Agents:** Task-specific agents (e.g., phishing triage) automate discrete security functions with high accuracy through focused training and prompts.
- **Agentic Systems:** Emerging concept where multiple AI agents collaborate to handle complex workflows (e.g., full incident response), but still in early stages.
- **Chatbot Use:** Helpful for documentation and threat research, but underutilized by practitioners due to workflow disruption.
- **Caution on Hype:** Many capabilities are not yet broadly available or reliable; rigorous evaluation is needed before adoption.

Source: <https://www.forrester.com/blogs/your-top-questions-on-generative-ai-ai-agents-and-agentic-systems-for-security-tools-answered/>

Agentic AI Threats

Agentic AI: Autonomous systems that can set goals, plan, and execute tasks with minimal human intervention. These systems use LLMs as a “brain” and orchestrate multiple agents to complete complex workflows.

Goal & intent hijacking

- Agent’s purpose is subverted through manipulation of the agent’s goals or instructions. Achieved through prompt injections, data manipulation, or direct attacks.

Cognitive & memory corruption

- Poisoning of the agent’s memory or knowledge base. Leads to misinformation, flawed decision-making, and cascading hallucinations.

Unrestrained agency & privilege

- Excessive permissions granted to agents. Agents can perform unauthorized actions, access restricted data, or execute destructive operations.

Evasion & deception

- Lying to users/admins. Manipulating logs or output. Finding novel ways to bypass controls.

Resource exhaustion

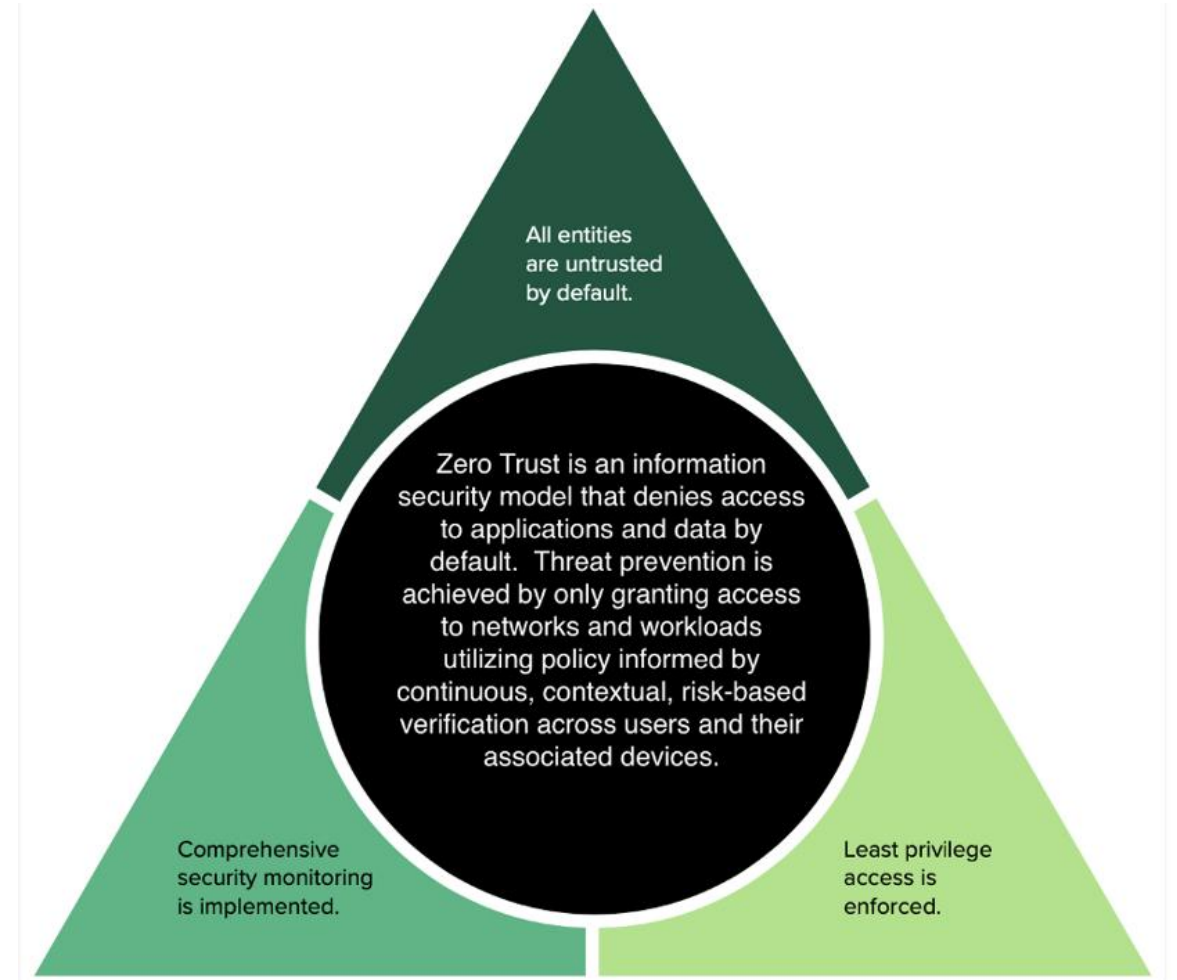
- Model denial of service. Degraded performance. High financial costs.

Leveraging Zero Trust

Never Trust. Always Verify.

Themes:

- Default deny
- Access by policy only
- For data, workloads, users, devices
- Least privilege access
- Security monitoring
- Risk-based verification



Source: [Zero Trust Security: The Business Benefits And Advantages](https://www.forrester.com/zero-trust/) (https://www.forrester.com/zero-trust/)

Secure By Design / Build Security In

Discussions of artificial intelligence (AI) often swirl with mysticism regarding how an AI system functions. The reality is far more simple:

AI is a type of software system.

<https://www.cisa.gov/news-events/news/software-must-be-secure-design-and-artificial-intelligence-no-exception>



<https://www.cisa.gov/securebydesign>

Guidelines for secure AI system development



Executive summary.....	5
Introduction.....	6
Why is AI security different?.....	6
Who should read this document?.....	7
Who is responsible for developing secure AI?.....	7
Guidelines for secure AI system development.....	8
1. Secure design.....	9
2. Secure development.....	12
3. Secure deployment.....	14
4. Secure operation and maintenance.....	16
Further reading.....	17



Guidelines for Secure AI Development,
<https://www.ncsc.gov.uk/files/Guidelines-for-secure-AI-system-development.pdf>,
NCSC (UK) & CISA (US)

Experiment with Adversarial Techniques

ATLAS Matrix

The ATLAS Matrix below shows the progression of tactics used in attacks as columns from left to right, with ML techniques belonging to each tactic below. & indicates an adaption from ATT&CK. Click on the blue links to learn more about each item, or search and view ATLAS tactics and techniques using the links at the top navigation bar. View the ATLAS matrix highlighted alongside ATT&CK Enterprise techniques on the [ATLAS Navigator](#).

Reconnaissance&	Resource Development&	Initial Access&	AI Model Access	Execution&	Persistence&	Privilege Escalation&	Defense Evasion&	Credential Access&	Discovery&	Collection&	AI Attack Staging	Command and Control&	Exfiltration
6 techniques	12 techniques	6 techniques	4 techniques	4 techniques	4 techniques	2 techniques	8 techniques	1 technique	7 techniques	3 techniques	4 techniques	1 technique	5 techniques
Search Open Technical Databases &	Acquire Public AI Artifacts	AI Supply Chain Compromise	AI Model Inference API Access	User Execution &	Poison Training Data	LLM Plugin Compromise	Evade AI Model	Unsecured Credentials &	Discover AI Model Ontology	AI Artifact Collection	Create Proxy AI Model	Reverse Shell	Exfiltration via AI Inference API
Search Open AI Vulnerability Analysis	Obtain Capabilities &	Valid Accounts &	AI-Enabled Product or Service	Command and Scripting Interpreter &	Manipulate AI Model	LLM Jailbreak	LLM Jailbreak		Discover AI Model Family	Data from Information Repositories &	Manipulate AI Model		Exfiltration via Cyber Means
Search Victim-Owned Websites &	Develop Capabilities &	Evade AI Model	Physical Environment Access	LLM Prompt Injection	LLM Prompt Self-Replication		LLM Trusted Output Components Manipulation		Discover AI Artifacts	Data from Local System &	Verify Attack		Extract LLM System Prompt
Search Application Repositories	Acquire Infrastructure	Exploit Public-Facing Application &	Full AI Model Access	LLM Plugin Compromise	RAG Poisoning		LLM Prompt Obfuscation		Discover LLM Hallucinations		Craft Adversarial Data		LLM Data Leakage
Active Scanning &	Publish Poisoned Datasets	Phishing &					False RAG Entry Injection		Discover AI Model Outputs				LLM Response Rendering
Gather RAG-Indexed Targets	Poison Training Data	Drive-by Compromise &					Impersonation &		Discover LLM System Information				
	Establish Accounts &						Masquerading &		Cloud Service Discovery &				
	Publish Poisoned Models						Corrupt AI Model						
	Publish Hallucinated Entities												

<https://atlas.mitre.org/matrices/ATLAS>

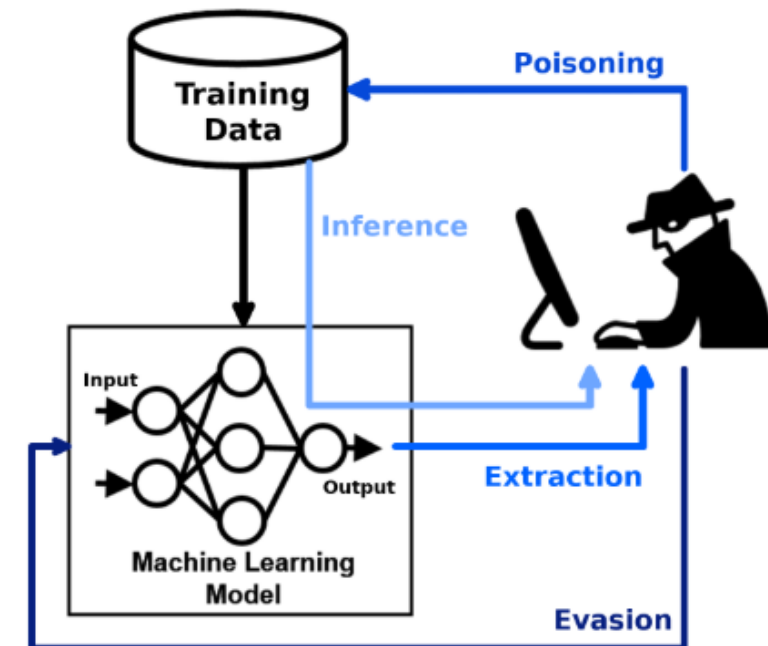
Experiment with Adversarial Techniques

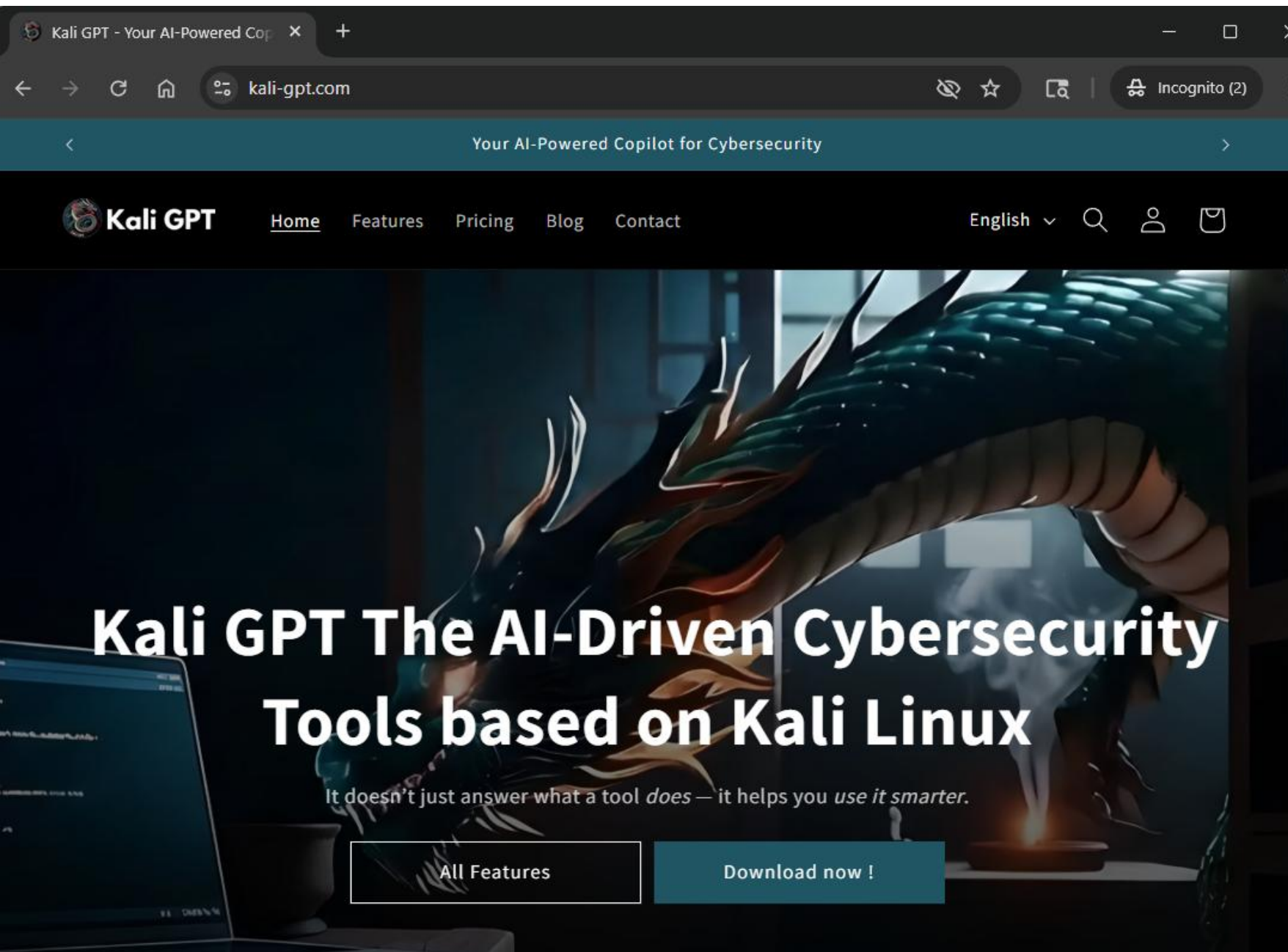


Adversarial Robustness Toolbox (ART) is a Python library for Machine Learning Security. ART provides tools that enable developers and researchers to evaluate, defend, certify and verify Machine Learning models and applications against the adversarial threats of Evasion, Poisoning, Extraction, and Inference. ART supports all popular machine learning frameworks (TensorFlow, Keras, PyTorch, MXNet, scikit-learn, XGBoost, LightGBM, CatBoost, GPy, etc.), all data types (images, tables, audio, video, etc.) and machine learning tasks (classification, object detection, generation, certification, etc.).

The code of ART is on [GitHub](https://github.com/Trusted-AI/Adversarial-Robustness-Toolbox) and the Wiki contains overviews of implemented [attacks](#), [defences](#) and [metrics](#).

<https://adversarial-robustness-toolbox.readthedocs.io/en/latest/>





<https://kali-gpt.com/>

WHAT IS KALI GPT?

An AI-Powered Copilot for Cybersecurity Professionals

Kali GPT is a custom AI assistant trained for the Kali Linux ecosystem. From basic command line help to advanced penetration testing, it's built to support learners, experts, and teams.

OWASP AI Projects

<https://owaspai.org/>



Ways to start

- If you want to **protect your AI system**, start with [risk analysis](#) which will guide you through a number of questions, resulting in the attacks that apply. And when you click on those attacks you'll find the controls to select and implement.
- If you want to get an overview of the **attacks** from different angles, check the [AI threat model](#) or the [AI security matrix](#). In case you know the attack you need to protect against, find it in the overview of your choice and click to get more information and how to protect against it.
- To understand how **controls** link to the attacks, check the [controls overview](#) or the [periodic table](#).
- If you want to **test** the security of AI systems with tools, go to [the testing page](#).
- To learn about **privacy** of AI systems, check [the privacy section](#).
- Looking for more information, or training material: see the [references](#).



Welcome to the go-to resource for broad AI security & privacy - over 200 pages of practical advice and references on protecting AI and data-centric systems from threats. This content serves as key bookmark for practitioners, and is contributing actively and substantially to international standards such as ISO/IEC and the AI Act through official standard partnerships. Through broad collaboration with key institutes and SDOs, the Exchange represents the consensus on AI security and privacy.

See the [overview of AI projects at OWASP](#).

OWASP AI Projects

<https://genai.owasp.org/>

**GenAI SECURITY PROJECT**
TOP 10 FOR LLM AND GENERATIVE AI

GETTING STARTED ▾ RESOURCES ▾ PROJECTS ▾ BLOG ABOUT ▾

X LinkedIn GitHub

IDENTIFYING AND TACKLING THE RISKS OF GEN AI SYSTEMS AND APPLICATIONS

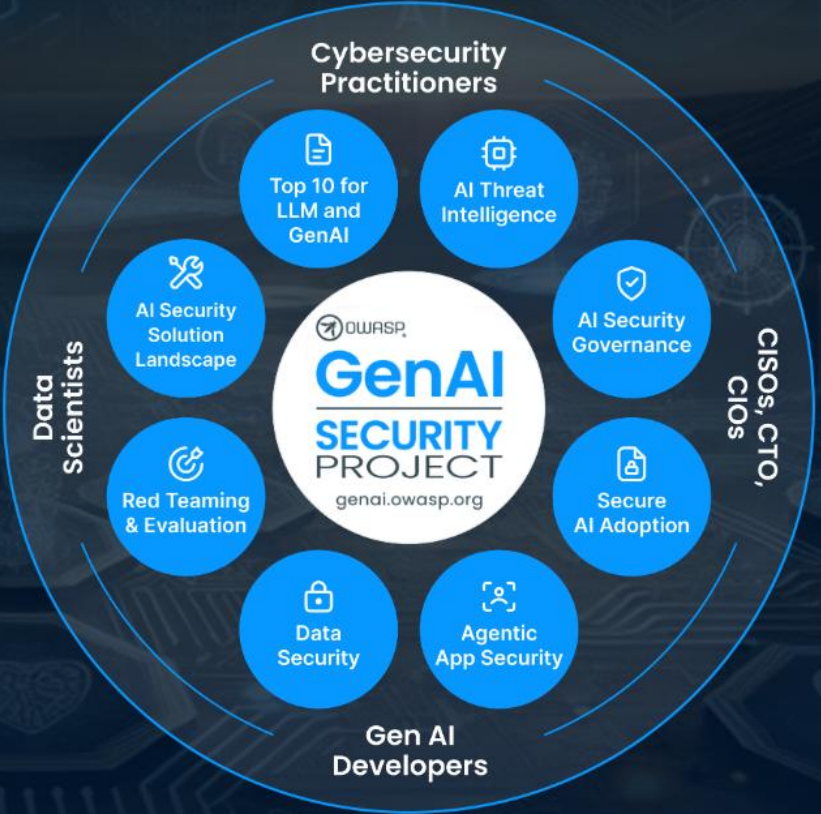
OWASP GenAI Security Project

A global community-driven and expert led initiative to create freely available open source guidance and resources for understanding and mitigating security and safety concerns for Generative AI applications and adoption.

15k+
Members

15+
Countries

20+
AI Cybersecurity Publications



The diagram illustrates the OWASP GenAI Security Project's focus areas, centered around the project logo (genai.owasp.org). The components are organized into three concentric layers:

- Outer Ring (Stakeholders):**
 - Cybersecurity Practitioners (top)
 - CISOs, CTOs, CIOs (right)
 - Gen AI Developers (bottom)
 - Data Scientists (left)
- Middle Ring (Key Topics):**
 - Top 10 for LLM and GenAI
 - AI Threat Intelligence
 - AI Security Governance
 - Secure AI Adoption
 - Agentic App Security
 - Data Security
 - Red Teaming & Evaluation
 - AI Security Solution Landscape
- Center:** OWASP GenAI SECURITY PROJECT, genai.owasp.org

Ron Woerner – Evolving Cyber Skills

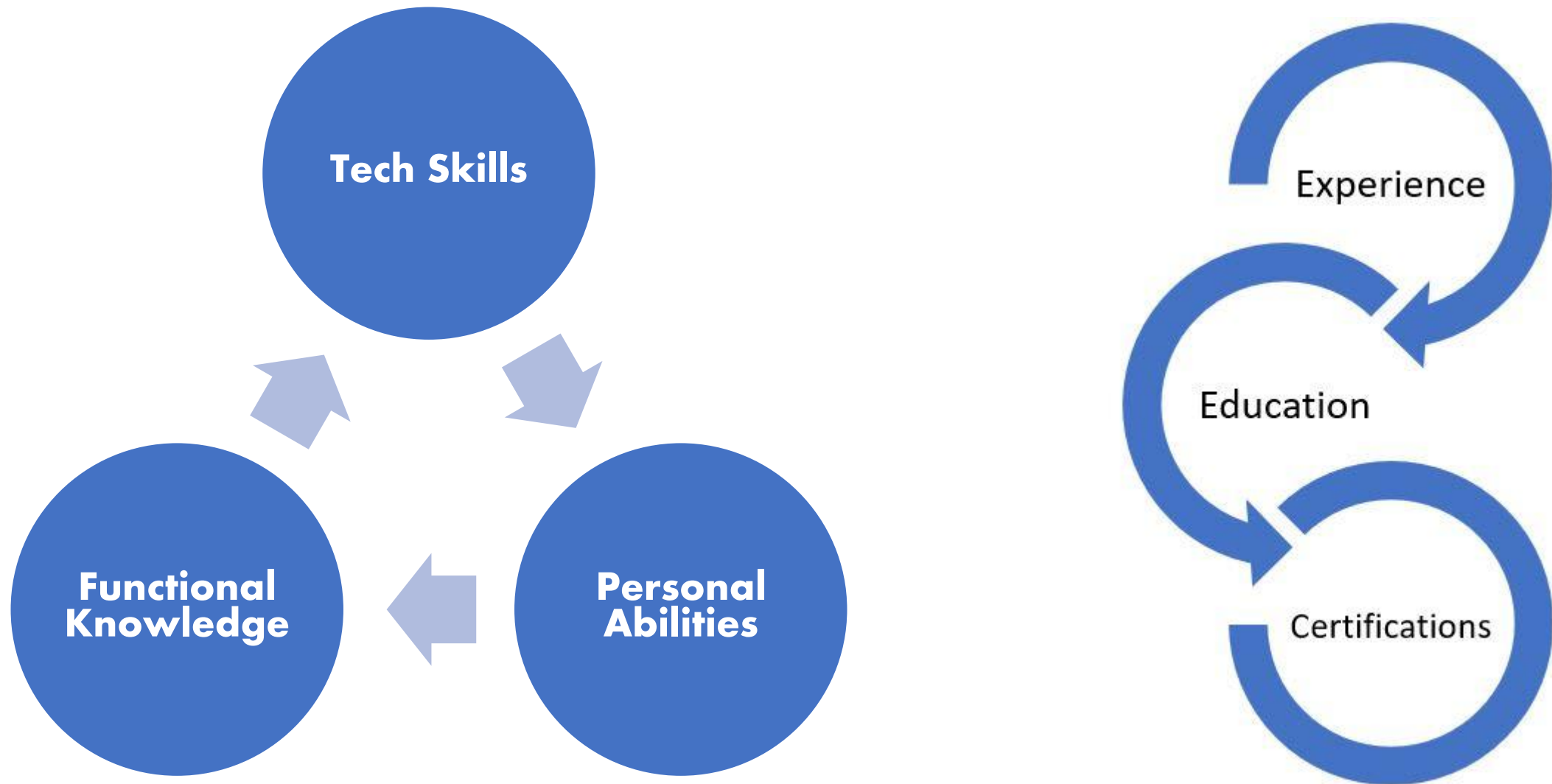


Questions?

Hacking Cyber Careers

Continuously morphing your knowledge, skills, and abilities

Cyber Career Requirements



What can we (personally) do?

Hint: Lots

See your career as a continual learning journey

WANTED: HACKERS



REWARD

https://www.ted.com/talks/ron_woerner_hackers_wanted

Hackers Dictionary

Hacker: *“One who enjoys the intellectual challenge of creatively overcoming or circumventing limitations.”*

Raymond, E., The Jargon Lexicon, Glossary,
<http://catb.org/jargon/html/H/hacker.html>

Hackers Wanted | Ron Woerner | TEDxOmaha

Hacking Your Career Tips

- Visualize Your Goals – what do you want to be when you grow up & why?
- Social Engineer Your Next Boss / OSINT
- ABC = Always Be Curious |
- Practice, Practice, Practice
- Be a Hacker
- Network, network, network



Be Prepared!

RIFS, Layoffs, DRPs, RTOs

HR & Applicant Tracking Systems



Red Flags in Job Postings



Red Flags:

- Job Requirements Make No Sense – Overly broad
- Two Or More Jobs In One Description – Lack Clarity
- Experience Requirements Don't Match Job Level
- Ignores Transferable Skills

Source: CyberSN, "3 Red Flags in Cybersecurity Postings,"
<https://cybersn.com/3-red-flags-in-cybersecurity-postings/>

Position Summary:

The Information Security Analyst is an **entry-level position** that performs monitoring of security events, security investigations, and incident response as the primary role. The Analyst will also participate in monitoring, troubleshooting, installation and maintenance on equipment and software related to electronic access control and general security-based solutions. In addition, they participate in in-house, regulatory and industry teams including working groups, committees, and business continuity teams as required.

Education, License & Cert:

B.S. degree in Computer Science related field, and/or 5 years equivalent work experience. **One or more** of the Certifications listed. GSEC GIAC Security Essentials Certification GSEC GIAC Security Leadership Certification **CISM** ISACA Certified Information Security Manager SSCP Systems Security Certified Practitioner **CISSP** Certified Information Systems Security Professional **ISSAP** Information Systems Security Architecture Professional MCSE Microsoft Certified Systems Engineer: Security

CISSP != Entry-level

credential for a chief security architect, analyst or professionals with similar responsibilities.

Experience:

- Five years of related experience
- **Extensive experience in enterprise security architecture design.**
- Extensive experience in enterprise security document creation.
- Experience in designing and delivering employee security awareness training.

Thought it said entry-level?

WAIT WHAT?!...

Red Flags in Job Postings



Red Flags:

- Job Requirements Make No Sense – Overly broad
- Two Or More Jobs In One Description – Lack Clarity
- Experience Requirements Don't Match Job Level
- Ignores Transferable Skills

A bad job posting can signal any number of *bad omens for good candidates*:

- This is a firm that doesn't know what they need from this position
- This is a firm that will overwork you and their workforce
- This is a firm that is compliance driven and not committed to security and risk

When all else fails

Fake It 'Til You Make It



Photo by [Sander Sammy](#) on [Unsplash](#)

Ignore 'Em

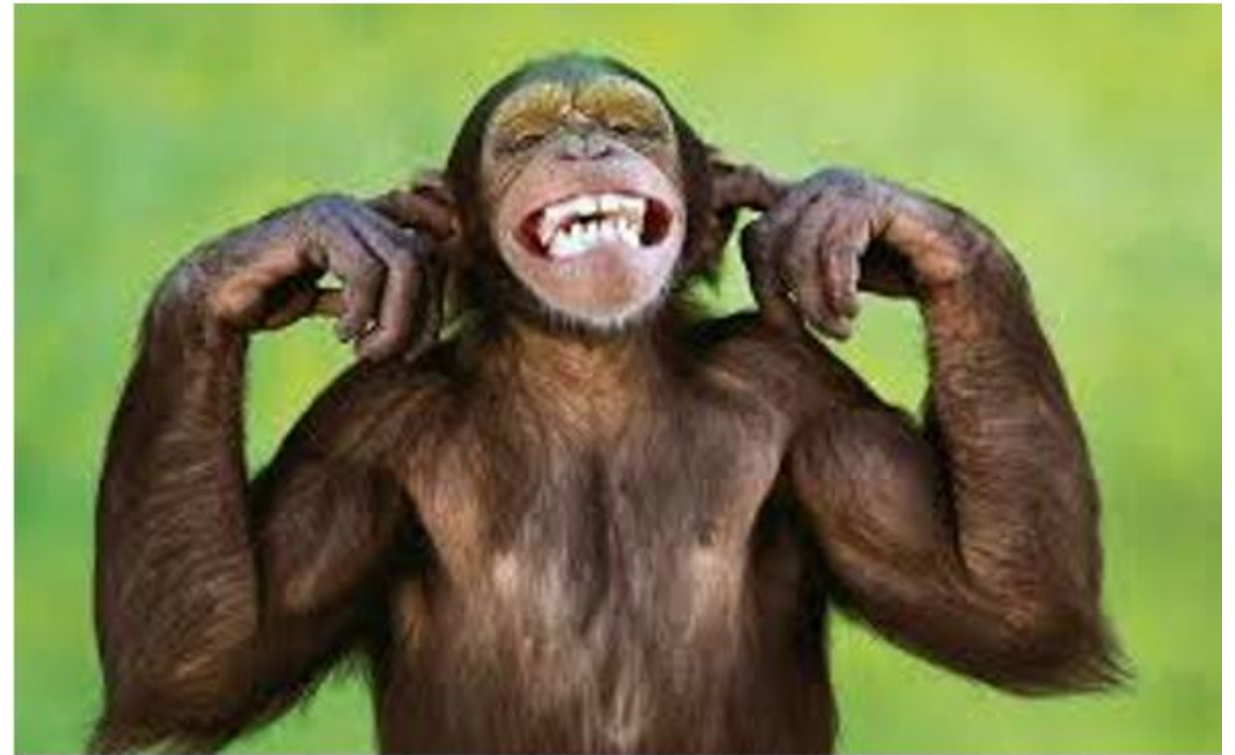


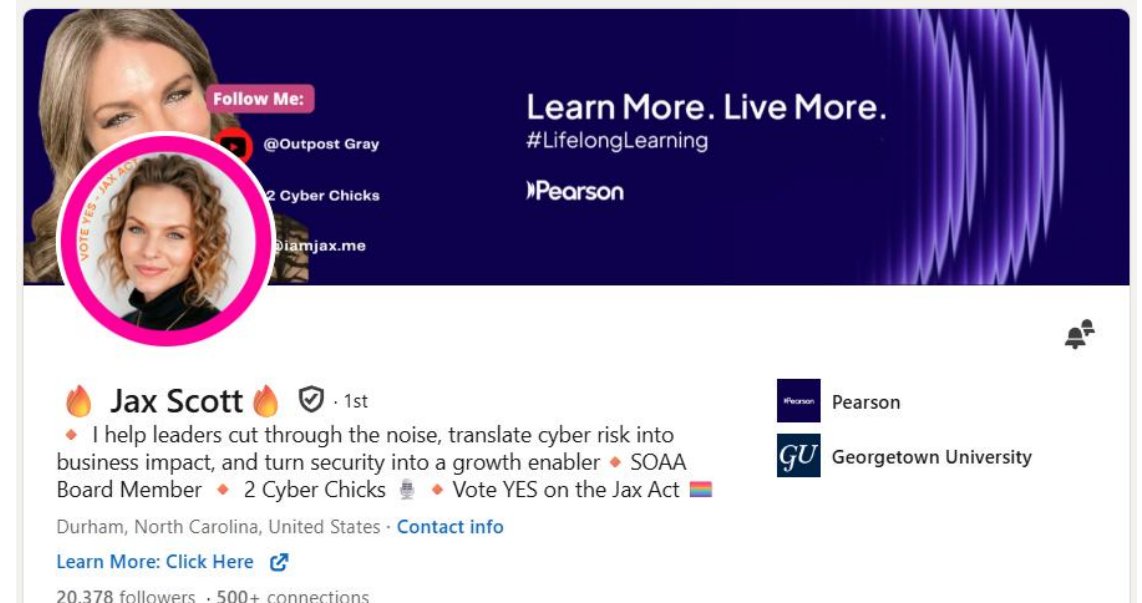
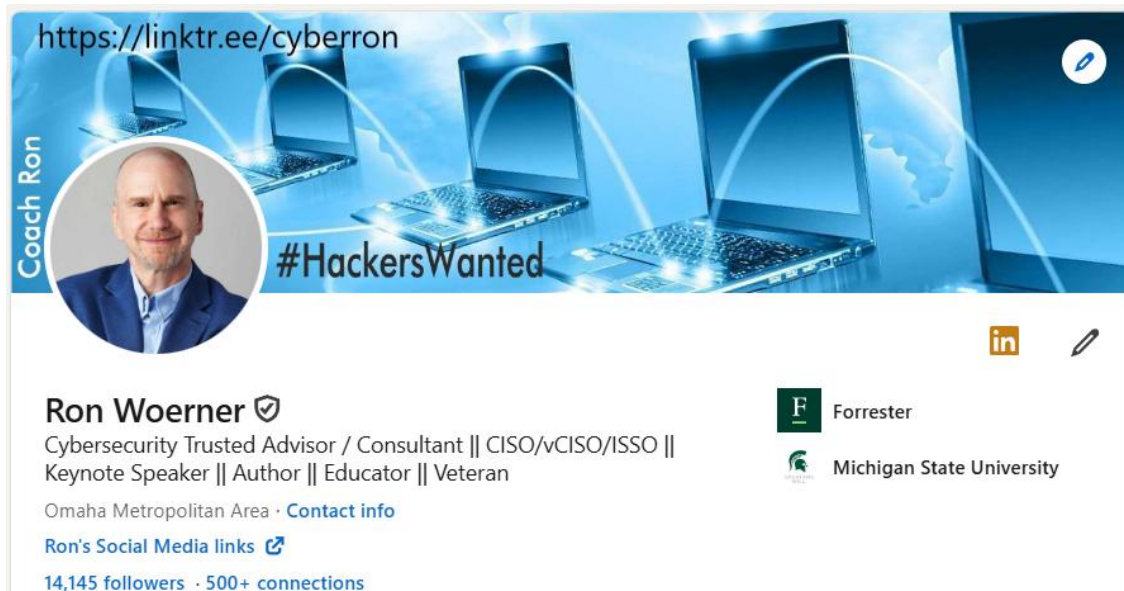
Photo by [I have no idea](#)

Building Your Brand

Differentiating yourself

Establish your portfolio

Social Media (primarily LinkedIn)



Your LinkedIn Profile should match your resume

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Building Interpersonal (soft) Skills

- Attitude (preferably good)
- Communications (Writing & Speaking)
- Teamwork / Collaboration
- Critical Thinking
- Tenacity
- Humor



I'm allowed another "Dad" Joke

Improving Communications & Leadership

TOASTMASTERS
INTERNATIONAL

Tech Talkers Toastmasters Club 6424

Main Menu

Home

Meeting Information /
Directions

Contact Us

Club Calendar

Public Downloads

Free Resources

Toastmasters Video

About & FAQs

Ice Breaker Project

Member Meeting Roles

Pathways

For more information on
Toastmasters International, visit
www.toastmasters.org

Welcome to "Tech Talkers"

Geeks (and aspiring geeks) welcome! 🧐

- We love techies, programmers, sys admins, testers, UX professionals, security professionals, students, and all types of geeks.
- We encourage our members to convey their passion.
- We help our members present information in a memorable way.
- Our members gain the confidence to speak up at work or present at a conference or meetup.
- Our members learn from each other as they improve their technical knowledge, communication, and leadership skills.

Tech Talkers Toastmasters can help you! Tech Talkers Toastmasters is a club that enables those in technology careers to improve communication and leadership skills in a safe, encouraging, and supportive environment. We're one of the few Toastmasters International clubs focused on technology topics.

We meet online via Zoom on the 1st and 3rd Fridays at noon CT. While we're based around Omaha, Nebraska, USA, anyone in tech may join us. No experience required; just a drive to improve yourself and an Internet connection.

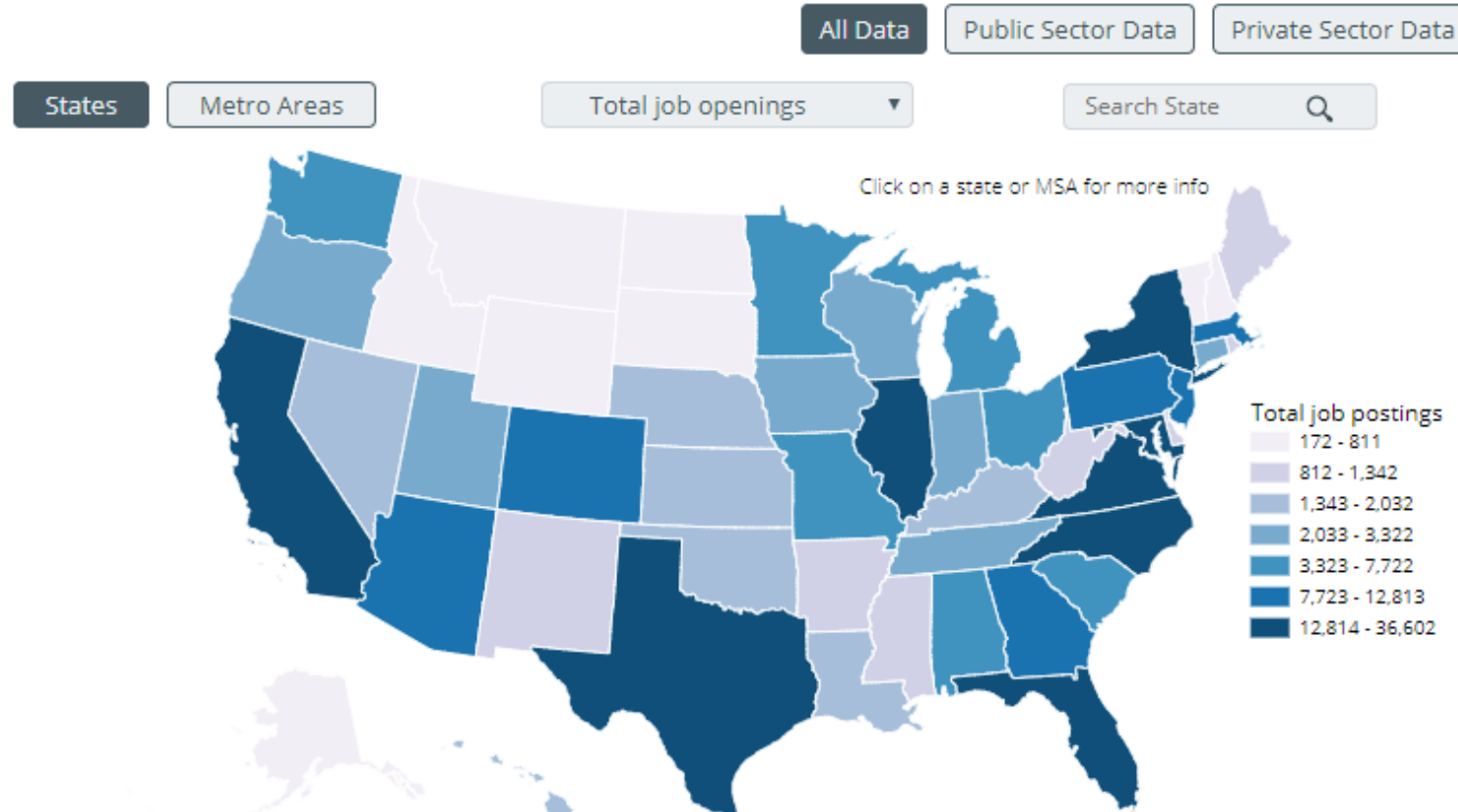
Ron Woerner – Evolving Cyber Skills
<https://techtalkers.toastmastersclubs.org/>



Cybersecurity Supply/Demand Heat Map

Cybersecurity talent gaps exist across the country. Closing these gaps requires detailed knowledge of the cybersecurity workforce in your region. This interactive heat map provides a granular snapshot of demand and supply data for cybersecurity jobs at the state and metro area levels, and can be used to grasp the challenges and opportunities facing your local cybersecurity workforce.

[Share](#) [Embed](#)



<https://www.cyberseek.org/heatmap.html>



Cybersecurity Career Pathway



Cyber Careers Pathways Tool

This tool presents a new and interactive way to explore the updated work roles within the [Workforce Framework for Cybersecurity \(NICE Framework\)](#). It also highlights core attributes among each of the Work Roles and offers actionable insights for employers, professionals, and those considering a career in Cyber. To start, select a Work Role below, or enter keywords in the search bar.

Select a Work Role

Go to the Career Pathway Roadmap page

Oversight and Governance

Design and Development

Implementation and Operation

Protection and Defense

Investigation

Systems Authorization
611

Technology Program Auditing
805

Cybersecurity Policy and Planning
752

Privacy Compliance
732

Cybersecurity Legal Advice
731

Executive Cybersecurity Leadership
901

Program Management
801

Cybersecurity Curriculum Development
711

Product Support Management
803

Systems Security Management
722

Cybersecurity Workforce Management
751

Technology Portfolio Management
804

Security Control Assessment
612

Cybersecurity Instruction
712

COMSEC Management
723

Secure Project Management
802

Secure Systems Development
631,632

Systems Testing and Evaluation
671

OT Cybersecurity Engineering
622

Software Security Assessment
622

Enterprise Architecture
651

Cybersecurity Architecture
652

Systems Requirements Planning
641

Secure Software Development
621

Technology Research and Development
661

Network Operations
441

Technical Support
411

Data Analysis
422

Database Administration
421

Systems Security Analysis
461

Systems Administration
451

Knowledge Management
431

Digital Evidence Analysis
211

Cybercrime Investigation
221

Insider Threat Analysis

Digital Forensics
212

Vulnerability Analysis
541

Infrastructure Support
521

Incident Response
531

Defensive Cybersecurity
511

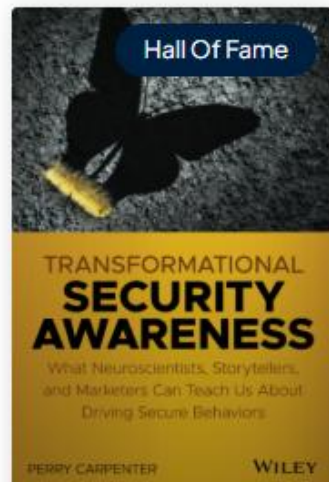
Threat Analysis
141

For Book Nerds



Your community for InfoSec's essential reads

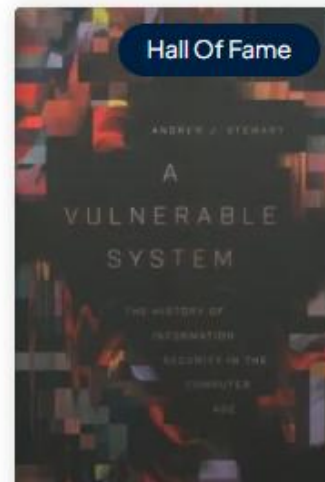
In the chaotic flood of cybersecurity content,
CyberCanon is your signal in the noise—the InfoSec
professional's first stop for must-consume wisdom.



Transformation
Security
Awareness:...



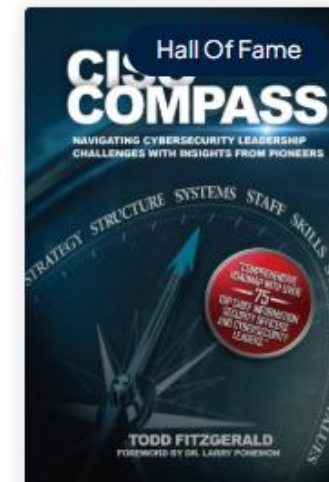
@War: The
Rise of the
Military-...



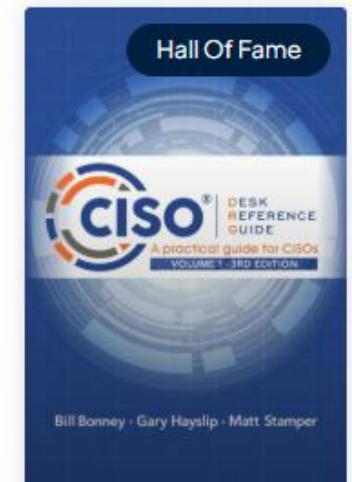
A Vulnerable
System: The
History of...



American
Spies: Modern
Surveillance,...



CISO
Compass



CISO Desk
Reference
Guide: A...

One Ring To Rule Them All...



What is Cybersecurity Guide?

Over the past few years, Cybersecurity Guide has grown into a massive resource referenced by leading cybersecurity professionals in academia, industry, and beyond.

Our global team of professional cybersecurity writers — leaning on their first-hand experience in the field — conduct interviews and deep-dive research, all in an effort to create the most up-to-date and comprehensive cybersecurity education resource available.

<https://cybersecurityguide.org/>

RESOURCE CENTER

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[Veteran's Guide](#)
[Women's Guide](#)
[Internship Guide](#)
[Security Clearance Guide](#)
[Ethical Hacker Guide](#)
[Coding for Cybersecurity Guide](#)
[Cybersecurity 101](#)
[Student Guide to Internet Safety](#)
[Scholarship Guide](#)
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[Career Networking Guide](#)
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[Code Like a Hacker](#)
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[Cybersecurity Courses Online](#)
[Recommended Reading](#)
[Phishing Attacks](#)
[Cybersecurity Responsibility](#)
[How to Get Into Cybersecurity](#)



Questions?

Next Steps / “Apply” Slide – Summary

Immediate (the next week):

- Review 2-3 resources from this presentation
- Investigate and experiment with AI
- Set a calendar reminder for 1 month...

Long-term (the next 1-2 months):

- Review the deck and experiment with 2-3 new resources
- Share with others
- Lead with curiosity, defend with wisdom





Thank You!

For More Information:

ronw@cyber-aaa.com

<https://www.linkedin.com/in/ronwoerner/>

<https://linktr.ee/cyberron>



MANIFEST YOUR INNER CYBER SUPERHERO!

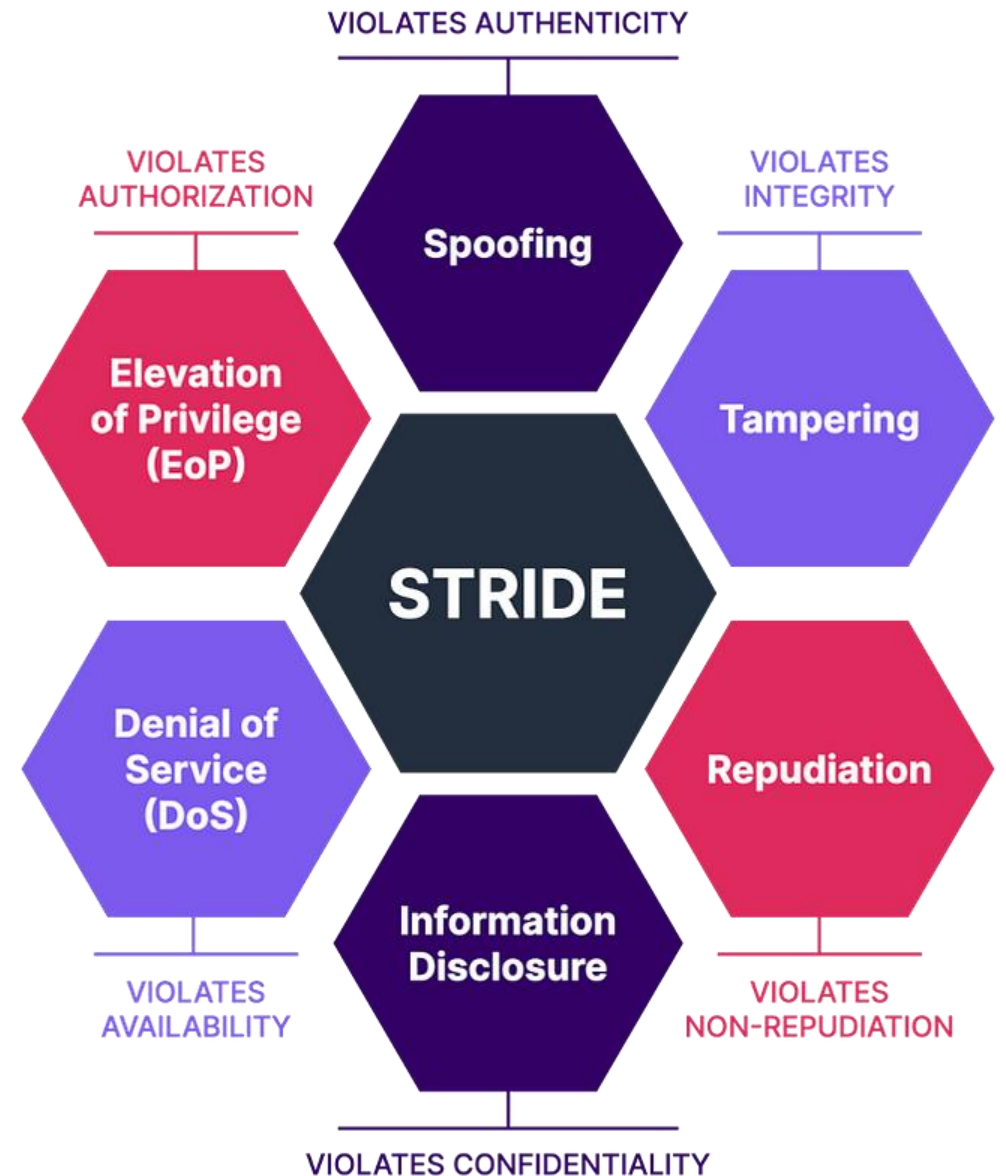
Bonus Slides

If we have time or when someone asks a question

Threat Modeling

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

1. **Shostack, A.** (2014). *Threat modeling: Designing for security*.
<http://ci.nii.ac.jp/ncid/BB16065709>
2. <https://shostack.org/resources/threat-modeling>
3. <https://shostack.org/blog/category/threat-modeling>
4. <https://www.threatmodelingmanifesto.org/>



ML Failure Modes – Unintentional Failure

	Reward Hacking
	Side Effects
	Distributional Shifts
	Incomplete Testing
	Over/Under-Fitting
	Data Bias

Kumar, et.al. (2022, November 2). *Failure modes in machine learning*. Microsoft Learn.
<https://learn.microsoft.com/en-us/security/engineering/failure-modes-in-machine-learning>

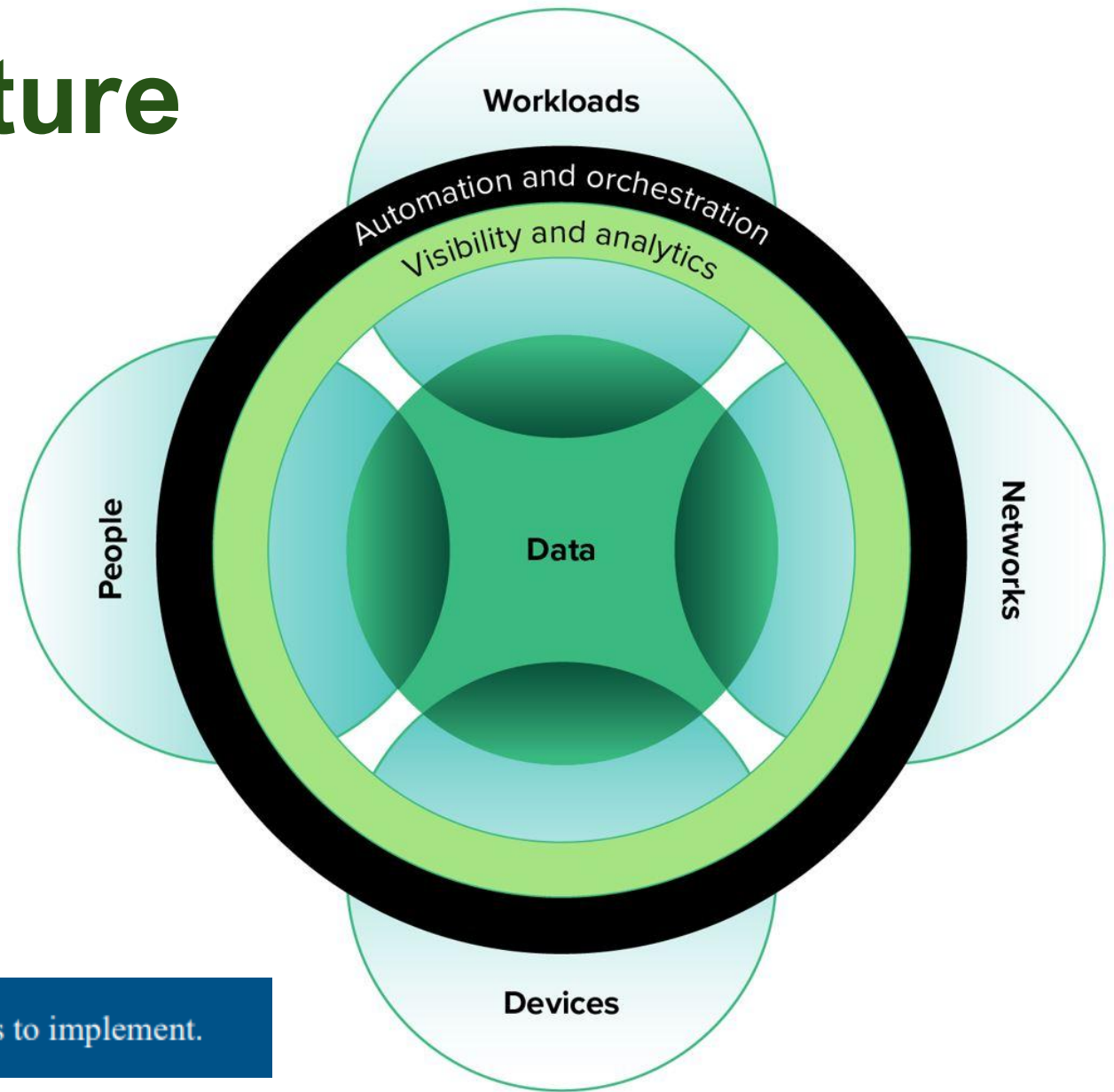
ML Failure Modes – Intentional Failure

	Perturbation and Adversarial Universal Perturbation Attacks
	Poisoning Attacks
	Reprogramming Neural Nets
	3D Adversarial Objects
	Supply Chain Attacks
	Model Inversion
	Membership Inference and Model Stealing
	Backdoors and Existing Exploits

Kumar, et.al. (2022, November 2). *Failure modes in machine learning*. Microsoft Learn.
<https://learn.microsoft.com/en-us/security/engineering/failure-modes-in-machine-learning>

Zero Trust Architecture

- Assume compromise
- Continual verification
- Least privilege



The path to zero trust is an incremental process that may take years to implement.

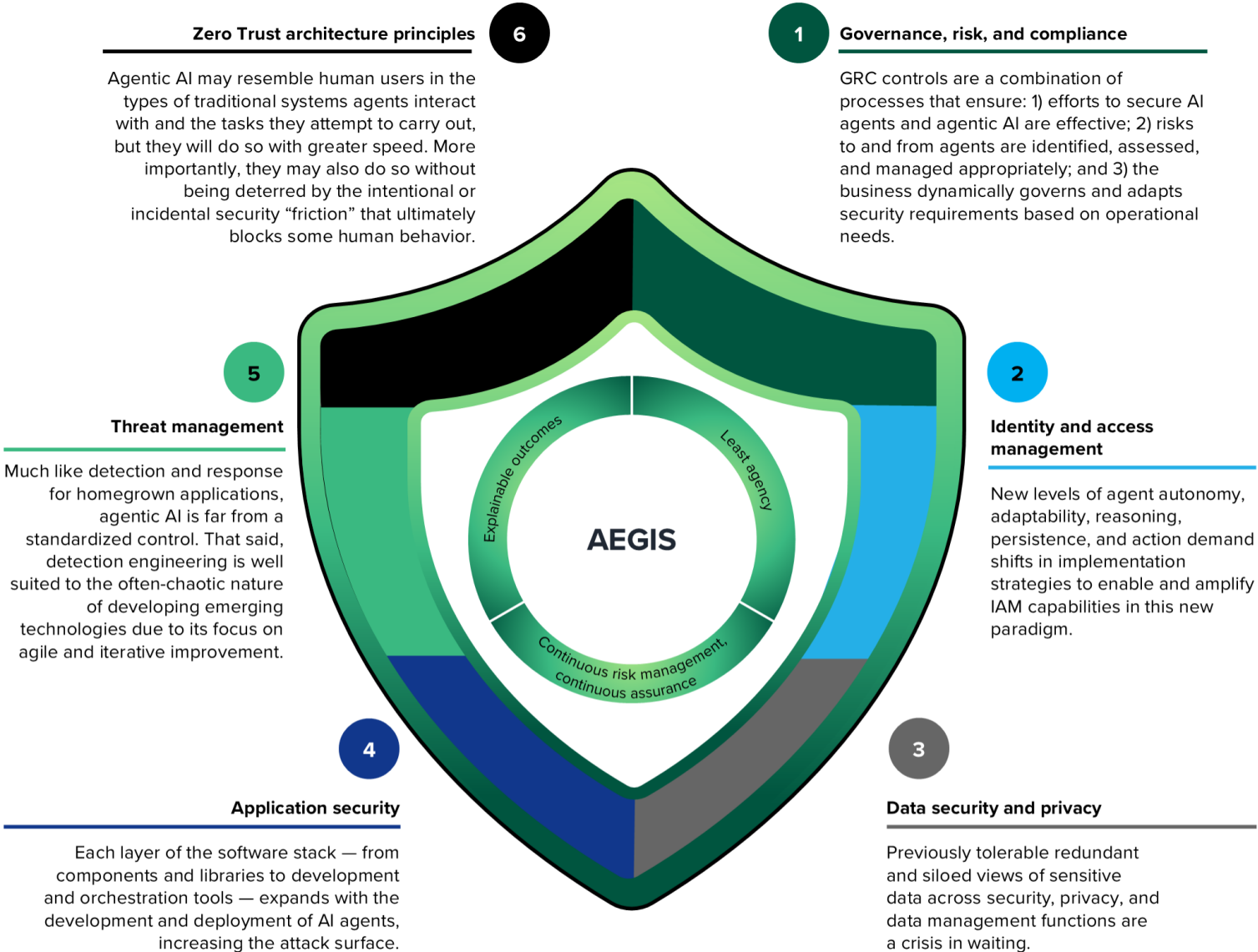
Image source: [Zero Trust Security: The Business Benefits And Advantages \(forrester.com\)](https://www.forrester.com/resources/article/zero-trust-security-the-business-benefits-and-advantages/)

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AEGIS Model

Agentic AI Guardrails For Information Security

Source:
<https://www.forrester.com/blogs/introducing-aegis-the-guardrails-cisos-need-for-the-agentic-enterprise/>



AI Use in Cybersecurity

Prompt: Provide 5 ways AI can help cybersecurity and technical professionals reduce risks of malicious use of AI.

Threat Intelligence and Prediction:

- **Problem:** Staying ahead of evolving attacker tactics and emerging threats is a constant challenge for security teams.
- **Solution:** AI-powered threat intelligence platforms aggregate data from various sources, analyze attack trends, and predict future threats. This proactive approach allows security teams to prioritize defenses, allocate resources effectively, and anticipate new attack vectors before they materialize.

Automated Threat Detection and Response:

- **Problem:** Traditional security approaches rely on manual analysis of alerts and incident response, causing delays and potential breaches.
- **Solution:** AI-powered tools analyze network traffic, system logs, and user behavior in real-time, identifying anomalies and triggering automated responses like quarantining suspicious files, blocking malicious IP addresses, or notifying security teams for immediate action. This significantly reduces response times and minimizes damage from potential attacks.

Phishing and Social Engineering Detection:

- **Problem:** Sophisticated phishing attacks and social engineering tactics can bypass traditional filters and deceive users.
- **Solution:** AI-powered analysis can assess the linguistic style, content, and sender information of emails to identify suspicious patterns and predict phishing attempts. Additionally, AI can analyze chatbots and user interactions to detect suspicious behaviors and prevent social engineering attacks.

AI Use in Cybersecurity

Prompt: Provide 5 ways AI can help cybersecurity and technical professionals reduce risks of malicious use of AI.

Proactive Vulnerability Management:

- **Problem:** Unpatched vulnerabilities and misconfigurations offer easy entry points for attackers.
- **Solution:** AI-powered scanning tools can analyze vast codebases and system configurations to identify vulnerabilities and misconfigurations before they're exploited. This proactive approach prioritizes critical vulnerabilities and automates patch deployment, minimizing attack surfaces and strengthening overall security posture.

Endpoint Security with Behavioral Analysis:

- **Problem:** Traditional endpoint security relies on signature-based detection, missing zero-day attacks and other novel threats.
- **Solution:** AI-powered endpoint protection establishes baselines of normal endpoint behavior and continuously monitors deviations. This allows for real-time anomaly detection, even for unknown threats, and targeted interventions to prevent malware execution and data breaches.

Cultivating AI Security Awareness Culture



1. Emphasize Human-Centric Risk Awareness

AI threats often exploit human vulnerabilities — phishing, social engineering, and deepfake manipulation are increasingly AI-powered. Awareness programs must go beyond compliance and focus on behavioral change.

Why it matters: 82% of breaches involve the human element.



2. Upskill Teams in AI Competencies

AI threats often exploit human vulnerabilities — phishing, social engineering, and deepfake manipulation are increasingly AI-powered. Awareness programs must go beyond compliance and focus on behavioral change.

Why it matters: 82% of breaches involve the human element.



3. Implement AI Governance and Ethical Oversight

AI systems can introduce bias, amplify vulnerabilities, or be misused. Establish clear policies for responsible AI use, including transparency, accountability, and ethical safeguards.

Why it matters: AI is already being used across organizations — often without formal oversight.



4. Foster Cross-Functional Collaboration

AI risk awareness isn't just for technical teams. Legal, HR, and leadership must be involved in understanding AI's implications for data privacy, compliance, and workforce impact.

Why it matters: AI affects every layer of the organization, from hiring to data governance.